Multimodal Product Search: Retrieval, Ranking, and System Optimization

Junhao Zhang
Department of ECE
junhaoz@andrew.cmu.edu

Abstract

Product search combines exact constraints, semantic similarity, and visual evidence, while relevance evaluation must distinguish unseen products from true zero-interaction cold start. This study evaluates a modular retrieval and ranking system on 164,900 products and 8,956 queries aligned from human-judged ESCI and SQID. Frozen SBERT/CLIP representations support train-calibrated, image-mask-aware FAISS retrieval and a 28-feature ListNet ranker. Grouped query splits, disjoint product roles, and a prediction-before-label final lock protect evaluation. On Warm Test queries, XGBoost improves NDCG@10 by 1.20% over BM25; LambdaMART and ListNet improve by 5.80% and 6.28% over XGBoost. ListNet's Cold-Test improvement is 1.69%, with uncertainty spanning zero. The original ranking targets remain unmet. Fusion improves Warm known-positive Recall@50 by 14.35% over SBERT; projection adaptation and controlled ranking extensions lack adoption evidence. Historical weak-label experiments are reported separately. Equivalent inference preparation preserves candidates and ranking outputs. Four repeated local CPU measurements average 102.16 ms for cached-vector retrieval and ranking, 164.03 ms with live encoding, and 184.52 ms for complete HTTP; pooled P95 values are 204.99, 337.10, and 370.08 ms. Hot and Broad 64-client workloads expose queueing timeouts and two early Broad-load terminations. The English demo supports natural-language search, while the evidence does not establish stable error-free 64-client operation. The results support calibrated retrieval and inference engineering, while exposing modest ranking gains and cold-generalization limits.

1 Introduction

Product search combines several kinds of evidence. A query may name an exact model, describe a use case, specify a color or capacity, or contain an informal expression that shares few words with a product title. Text retrieval provides a strong starting point, but lexical overlap alone cannot resolve all such cases. Images offer complementary evidence about appearance and product identity, yet they are not uniformly available and need not share the score distribution of text embeddings. A useful system must combine these signals without rewarding missing content or confusing a relevant accessory with the requested product.

Cold-product evaluation introduces a second difficulty: an unseen product ID is a property of an experimental split, whereas a real new product also has a temporal history and an exposure process. Relevance labels, query construction, and candidate coverage affect what can be concluded about either setting. The initial implementation used Amazon Reviews'23 metadata, synthetic queries, and rule-based or Qwen-generated relevance labels. Those experiments supported a working multimodal pipeline, but a subsequent human review exposed limited agreement with the weak labels. The project was consequently rebuilt around ESCI human relevance judgments and SQID multimodal representations [1, 2].

This report asks four questions. First, does combining frozen textual and visual representations improve catalog retrieval when missing images and score scales are handled explicitly? Second, do pairwise or listwise ranking methods improve on a pointwise baseline under the same judged-pool evaluation? Third, can projection-only contrastive adaptation or targeted ranking changes improve generalization to products excluded from supervised training? Fourth, can the selected frozen system be served efficiently without altering its predictions? The completed work provides the following contributions and findings:

- A reproducible ESCI–SQID benchmark with frozen query and product roles, explicit target masks, image-coverage diagnostics, and a one-time final prediction lock.
- A calibrated, mask-aware fusion retriever and a controlled 28-feature ranking comparison. Final catalog known-positive recall improves over SBERT, while the original ranking improvement targets remain unmet.
- A documented sequence of adaptation, loss, feature, and retrieval experiments, including negative results and a separately registered RRF alternative.
- An inference implementation that preserves the selected models, feature definitions, and ranking rules, together with measured core, live HTTP, and bounded-concurrency performance.

The work integrates established methods rather than introducing a new ranking algorithm. The final model is a frozen modular pipeline; it is not jointly trained end to end. Historical metadata experiments remain part of the development record, but their scores are reported separately from the human-judged benchmark.

2 Related Work

Lexical and dense retrieval. BM25 models term-frequency saturation and document-length normalization and remains a useful lexical baseline [3]. Sentence-BERT enables independent query and document encoding for efficient similarity search [4]. This study uses the released all-MiniLM-L6-v2 checkpoint [5]; it does not train a new sentence encoder. Sparse and dense retrieval are evaluated over the same product catalog, rather than assuming that semantic similarity necessarily dominates exact matching.

Multimodal product representations. CLIP learns compatible image and text representations from natural-language supervision [6]. SQID enriches ESCI with product images and precomputed representations, making multimodal product-search experiments possible without downloading and re-encoding every image [2]. ESCI supplies Exact, Substitute, Complement, and Irrelevant judgments [1]. These are human relevance judgments over sampled query–product pairs, not exhaustive labels for every product in a catalog. The benchmark uses a custom split and gain mapping; its results are not official ESCI leaderboard scores. Its SBERT checkpoint also differs from the MiniLM-L12 baseline described by SQID.

Learning to rank. XGBoost provides scalable boosted-tree learning [7]. LambdaMART uses ranking-sensitive pairwise gradients, with efficient tree implementations available in LightGBM [8, 9]. ListNet instead matches distributions over complete query lists [10]. The experiments compare these established objectives using frozen training views and common features. Its gain-aware and position-aware ListNet alternatives are controlled experiments; neither is assumed superior because its loss resembles an evaluation metric.

Adaptation and retrieval combination. Contrastive objectives encourage paired representations to agree relative to in-batch alternatives [6]. The adaptation study here freezes CLIP and learns only small linear projection heads on permitted Warm product pairs. Reciprocal rank fusion combines ranked lists without requiring their raw scores to share a scale [11]. RRF60 is evaluated with equal Top100 budgets and remains an auxiliary system. Exact inner-product search is implemented with FAISS [12]; the present index is FlatIP, so the project does not claim an approximate-nearest-neighbor algorithmic contribution.

Historical query understanding. Earlier project versions explored LLM-based query processing, BGE representations and reranking, and CLIP LoRA adaptation [13]. These belong to the historical

metadata experiments. Query rewrite is disabled in the final ESCI system, and none of those additional rerankers or adapters is a dependency of the selected formal pipeline.

3 Methodology

3.1 Problem Formulation and Architecture

Let q be a query and $p \in \mathcal{P}$ a product in a fixed catalog. Product content comprises title, brand, bullet text, description, color, and an image-availability indicator m_p. Human relevance is observed only for a set $\mathcal{J}_q \subset \mathcal{P}$. Two tasks are kept separate: ranking the fixed judged candidate pool $\mathcal{J}_q$, and retrieving candidates from the complete catalog $\mathcal{P}$. An unjudged product is unknown, not an observed negative.

Figure 1 shows the final system. The retriever returns exactly 100 candidates from calibrated frozen embeddings. The ranker scores those candidates using the original feature contract and scaler. Training and evaluation may use judgments, but the inference service receives only query text and product content. The RRF alternative changes candidate retrieval only; it uses the same frozen ranker.

3.2 Frozen Representations and Exact Retrieval

SBERT provides normalized query and product vectors $s_q, s_p \in \mathbb{R}^{384}$. The product document concatenates the five available text fields. SQID provides L2-normalized CLIP product-title text, image, and query vectors $t_p, i_p, c_q \in \mathbb{R}^{768}$. Their inner products yield $s = s_q^\top s_p$, $t = c_q^\top t_p$, and $i = c_q^\top i_p$. A missing image contributes no image similarity; its product remains in the catalog.

The original late-fusion baseline combines SBERT with a CLIP text/image mixture, replacing the latter with text alone when an image is missing. This exposes a scale problem: CLIP text and

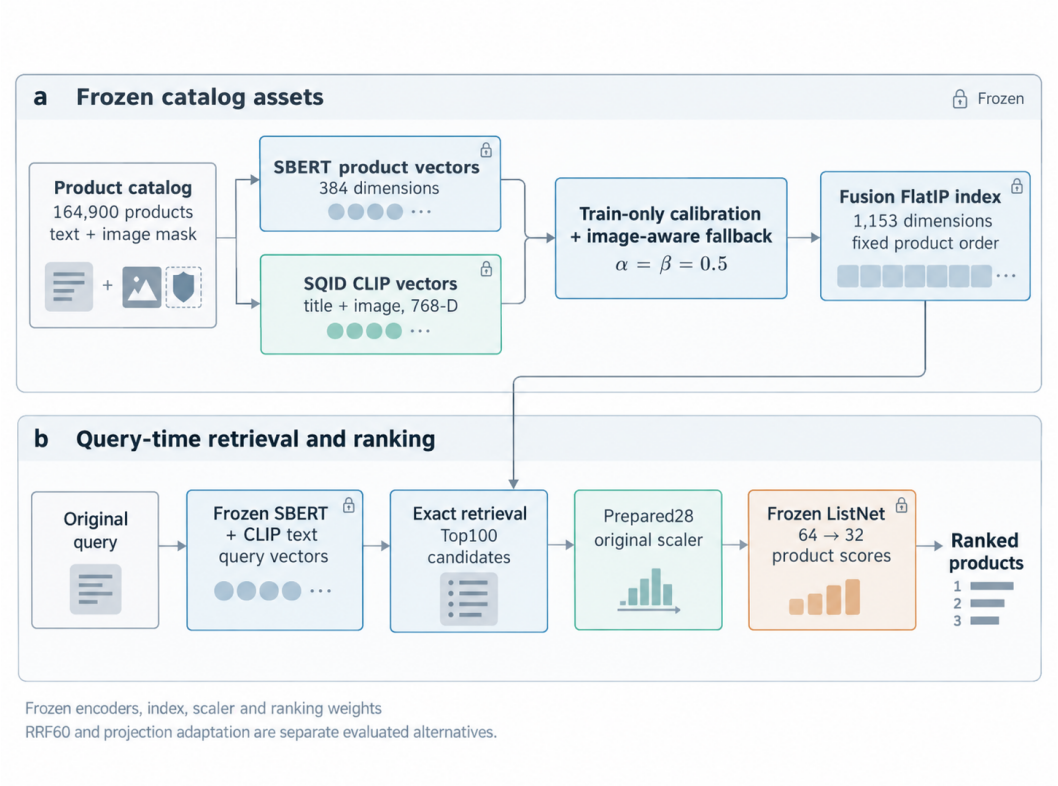


Figure 1: System architecture. (a) Frozen catalog representations and the calibrated exact-search index. (b) Query-time encoding, Top100 retrieval, original feature construction and frozen ListNet. RRF60 and projection adaptation are evaluated separately and are not default components. Cached benchmark vectors and live encoding retain distinct validation boundaries.

image cosines have different distributions, and the missing-image text distribution differs again. Replacing an image mixture with raw text cosine can therefore create an availability-dependent ranking advantage. Calibration estimates fixed moments from unlabeled Train-query/Warm-product scores, without relevance stratification or Validation/Test labels.

Let (μ_t, σ_t) and (μ_i, σ_i) be text and image moments for image-available pairs, (μ_m, σ_m) the missing-image text moments, and ρ the text–image correlation. Define

$$z_t = (t - \mu_t)/\sigma_t, \qquad z_i = (i - \mu_i)/\sigma_i, \tag{1}$$

$$v_\beta = \sqrt{(1-\beta)^2 + \beta^2 + 2\beta(1-\beta)\rho}, \tag{2}$$

$$h_\beta(q,p) = \begin{cases} (1-\beta)z_t + \beta z_i, & m_p = 1, \\ v_\beta(t - \mu_m)/\sigma_m, & m_p = 0, \end{cases} \tag{3}$$

$$C_\beta(q,p) = \mu_t + \sigma_t h_\beta(q,p), \qquad F(q,p) = \alpha s + (1-\alpha)C_\beta(q,p). \tag{4}$$

The missing-image branch matches the mixture's estimated scale rather than substituting a higher-scale text score. Warm Validation selected $\alpha = \beta = 0.5$; those weights were subsequently frozen. The calibration itself uses only Train score distributions.

Because C_β is affine in text and image similarity for each availability branch, F can be represented by one inner product. With branch-specific coefficients a_p, b_p, d_p satisfying $C_\beta = a_p t + b_p i + d_p$, store

$$u_p = [\alpha s_p;\ (1-\alpha)(a_p t_p + b_p i_p);\ (1-\alpha)d_p], \qquad u_q = [s_q;\ c_q;\ 1]. \tag{5}$$

This produces a 1,153-dimensional FlatIP index. The fused vectors are not renormalized after concatenation, because that would change the score. Exact search preserves the original index row order and uses deterministic product-key tie breaking, including ties at the Top100 boundary.

The auxiliary RRF route takes Top100 from each frozen BM25, SBERT, and calibrated fusion retriever. Products are deduplicated by their canonical key, assigned $\sum_r 1/(60 + \mathrm{rank}_r(p))$ over routes containing them, and truncated to Top100 before the same ListNet reranking. The larger raw union is used only as a coverage upper bound, not as an equal-budget competitor.

3.3 Features and Ranking Objectives

The Week2 pointwise baseline uses 20 lexical and metadata features: BM25, TF–IDF, title TF–IDF, token coverage in individual fields, digit and phrase matches, lengths, and missing-field indicators. Week5 adds eight features: SBERT cosine, CLIP text cosine, masked image cosine, image availability, a raw text-only fusion score, the calibrated fusion score, and brand/color lengths. There are no fabricated prices, ratings, review counts, or behavioral features. Appendix B specifies all 28 columns in execution order, including dtype and fallback details.

The original Week2 XGBoost predicts relevance grade by squared-error regression. The Week5 comparison retrains XGBoost, LambdaMART, and ListNet on the same eligible query lists and the same 28 columns. LambdaMART uses grades $0, 1, 2, 3$, gains $0, 1, 3, 7$, and an NDCG truncation level of 10. ListNet uses a $28 \to 64 \to 32 \to 1$ MLP with ReLU hidden layers. For query q, permitted training grades y_j and scores f_j define

$$\begin{aligned} P_y(j \mid q) &= \frac{e^{y_j}}{\sum_{k \in L_q} e^{y_k}}, \qquad P_f(j \mid q) = \frac{e^{f_j}}{\sum_{k \in L_q} e^{f_k}}, \\ \mathcal{L}_{\mathrm{LN}} &= -\frac{1}{|\mathcal{Q}|} \sum_q \sum_{j \in L_q} P_y(j \mid q) \log P_f(j \mid q). \end{aligned} \tag{6}$$

Here $\mathcal{Q}$ is the training query set, L_q contains the permitted Warm training items, and padding is masked before normalization. Query losses receive equal weight. The scaler is fitted only on training features: means and standard deviations are computed in float64, standardized values are then cast to float32, and near-zero standard deviations are replaced by one. All deployed weights and scaler parameters remain frozen.

3.4 Evaluated Adaptation and Controlled Extensions

The cold-adaptation experiment learns two bias-free 768×768 projection matrices, initialized to identity, while freezing the underlying CLIP vectors. Product text and queries share W_t; product

images use W_i. Outputs are L2-normalized. For a batch of B permitted product pairs, $T_j = \mathrm{norm}(W_t t_j)$ and $I_j = \mathrm{norm}(W_i i_j)$, with symmetric InfoNCE

$$\mathcal{L}_{\text{align}} = -\frac{1}{2B}\sum_{j=1}^{B}\left[\log\frac{e^{T_j^\top I_j/\tau}}{\sum_k e^{T_j^\top I_k/\tau}} + \log\frac{e^{I_j^\top T_j/\tau}}{\sum_k e^{I_j^\top T_k/\tau}}\right], \quad \tau = 0.07. \tag{7}$$

Only Warm-Train product pairs enter training; held-out Warm alignment pairs and both Cold roles are excluded. Adapted features are tested with the original frozen ListNet, using only Train-derived score calibration when required. This adapter is an evaluated alternative, not a component of the final system.

Subsequent controlled studies test query-balanced and hard-negative weights, tree regularization, five constraint features, a gain-aware ListNet target, a position-aware pair loss, and field-level matching. Gain-aware supervision fixes $\epsilon = 0.1$ in $P_y(j) \propto 2^{y_j} - 1 + \epsilon$. The later position-aware loss weights pairwise logistic terms by their current-rank $\Delta\mathrm{NDCG@10}$, with zero discount beyond rank 10. Architecture, folds, budgets, and comparisons are fixed in advance; these alternatives do not authorize changing the formal pipeline. Details and negative outcomes appear in Section 5.5 and Appendix D.

3.5 Inference and Service Implementation

The optimized implementation prepares product-static text, tokens, and lexical statistics once and computes candidate features in batches. A NumPy product-key array is aligned exactly with FAISS rows, replacing per-request selection and conversion of complete catalog DataFrames. Models, index, scaler, embeddings, and ID maps load at startup. These changes preserve feature mathematics, float-conversion points, Top100 retrieval, and deterministic ranking; they do not approximate the model.

A modular SearchService connects query encoding, fusion retrieval, feature construction, ranking, and real catalog display. FastAPI exposes health, readiness, validated natural-language search, and controlled replay endpoints; Streamlit calls the API rather than maintaining a second ranking implementation. Live SBERT/CLIP encoding was enabled after a separate vector-and-result compatibility study, with the unresolved historical CLIP revision recorded explicitly. A bounded queue serializes core computation, limits waiting, and returns explicit overload errors. A finite product-token cache reduces repeated static preparation; it is not a precomputed answer table for user queries.

4 Experimental Setup

4.1 Dataset Alignment and Available Content

The principal benchmark selects the US portion of the original ESCI Small Test data and aligns its product and query keys with SQID. It then constructs new project Train/Validation/Test roles. Thus, "Final Test" below refers to a project-held-out partition, not an untouched official ESCI challenge test. The aligned data contain 164,900 products, 8,956 queries, and 181,701 human-judged pairs. SQID provides valid image vectors for 156,100 products (94.66%); 8,800 products have missing or invalid image representations, including 442 identified placeholder cases. All products retain usable text representations. Missing images do not cause products or judgments to be dropped.

Product content is restricted to title, brand, bullet text, description, and color. The system does not infer missing commercial attributes. ESCI grades map E/S/C/I to $3/2/1/0$. All training and test relevance in this benchmark comes from the released human judgments, replacing the earlier mixture of rule-based training labels and Qwen silver judgments. A product's original corpus membership does not establish that it was absent from encoder pretraining.

4.2 Frozen Roles and Isolation

Query groups are allocated 70/15/15 to Train, Validation, and Test, producing 6,269/1,344/1,343 queries. Grouping uses normalized query text, token sets, and a fixed near-duplicate rule. Product roles are allocated 80/10/10 to Warm, Cold-Val, and Cold-Test, producing 131,920/16,490/16,490 IDs. Product allocation is stratified by occurrence and image availability; the seed is 20261014. These roles remain fixed throughout the ESCI experiments. The original Week2 supervised view uses 101,984 Train-query/Warm-product pairs. Listwise eligibility gives 6,171 queries and 100,856 pairs for Week5. The two views are preserved rather than silently equated.

Table 1: Dataset coverage, frozen roles, and training eligibility.

Population	Count	Role / boundary
Custom human-judged ESCI Small US Test rebuild		
Catalog products	164,900	Human ESCI judgments
Queries	8,956	Human ESCI judgments
Judged query–product pairs	181,701	Human ESCI judgments
Valid / unavailable image vectors	156,100 / 8,800	94.66% valid
Frozen independent query and product roles		
Query / Train	6,269	Grouped query split
Query / Validation	1,344	Grouped query split
Query / Final Test	1,343	Grouped query split
Product / Warm	131,920	Stratified product split
Product / Cold-Val	16,490	Stratified product split
Product / Cold-Test	16,490	Stratified product split
Training views (Train queries and Warm products only)		
Original pointwise	6,269 queries	101,984 pairs
Controlled LTR	6,171 queries	100,856 pairs
Projection training	88,063 products	Content-deduplicated Warm Train
Alignment validation	16,853 products	Disjoint paired-product gallery

Notes. Seed 20261014. Valid-image coverage is frozen embedding coverage. Query and product roles are independent. Pointwise and LTR eligibility differ; evaluation retains all candidate competitors.

Warm identifies a product partition, not actual supervised exposure: only 94,768 Warm products occur in Week2 supervision and 93,788 in Week5. No Cold product enters these supervised training views. Projection adaptation has an additional Warm-Train/Warm-Val content split, with 88,063 and 16,853 image–text pairs after the recorded content filters. Warm-Val alignment pairs are excluded from projection training. Content duplicates can nevertheless cross the general product roles: the audit finds 399 normalized-title, 1,020 image-URL, and 354 exact image-vector groups spanning partitions. Content-unique diagnostics therefore supplement ID-based Cold results.

Warm Validation selects the original retrieval and ranking configurations. Cold-Val is used only in the explicitly approved Week6 adaptation and later fixed safety comparisons; it is not used to select the subsequent Train-only ranking experiments. Those experiments use either a fixed internal 80/20 query-group split or three grouped folds. Pseudo-Cold experiments additionally remove held-out Warm product IDs from supervised training while retaining their content for inference and retaining all candidate competition positions.

The one-time final execution first generated all preregistered predictions for 1,343 Test queries and saved a content-addressed prediction lock. Only then were Test judgments for Warm and Cold-Test products unlocked. The authorized evaluation has 24,812 labels: 22,042 Warm and 2,770 Cold-Test. A further 2,675 Cold-Val candidate positions remain competitors with masked relevance. Neither this report nor the serving work performs a new final evaluation or changes a model in response to final scores.

4.3 Metrics, Scope, and Uncertainty

Let Overall target scope be Warm ∪ Cold-Test; Cold-Val remains excluded. For target scope S and query q, let $y_{q,p}$ be an observed grade, $G(y) = 2^y - 1$, and $\mathcal{P}_q^S = \{p \in \mathcal{J}_q \cap S : y_{q,p} \geq 2\}$. A scope is eligible when $|\mathcal{P}_q^S| > 0$. The fixed-pool ranking retains the complete candidate list, with gain masked outside S:

$$\mathrm{DCG}_{q,S}@10 = \sum_{r=1}^{\min(10,|\mathcal{J}_q|)} \frac{\mathbb{1}[p_r \in S] G(y_{q,p_r})}{\log_2(r+1)}, \qquad \mathrm{NDCG}_{q,S}@10 = \frac{\mathrm{DCG}_{q,S}@10}{\mathrm{IDCG}_{q,S}@10}. \quad (8)$$

IDCG sorts the target gains, not the whole unmasked list. MAP@10 uses E/S as positives and denominator $\min(10, |\mathcal{P}_q^S|)$; MRR is the reciprocal rank of the first target E/S item over the complete list, without a rank-10 cutoff. These are project metrics using gains $7/3/1/0$, distinct from the

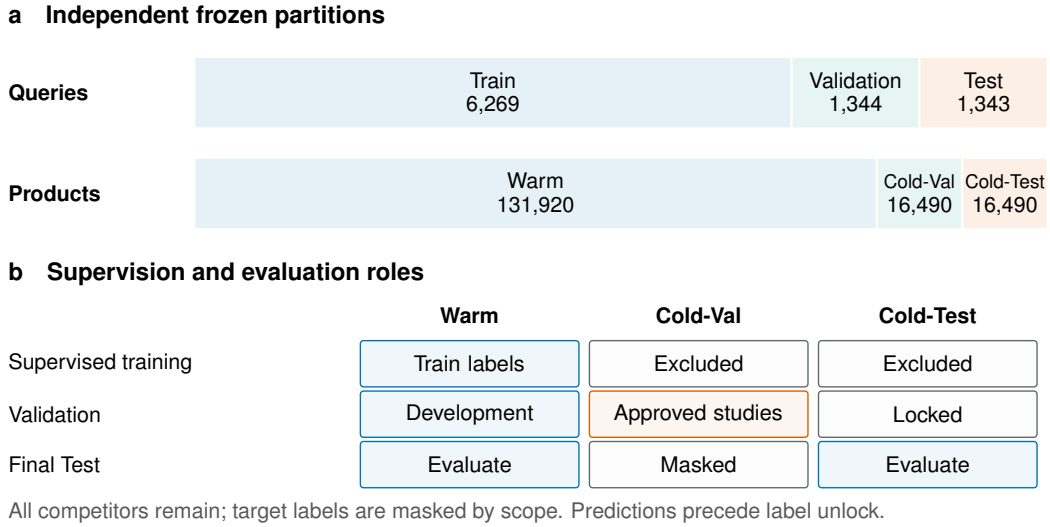


Figure 2: Frozen data protocol. Query partitions and product partitions are independent. The role matrix distinguishes permitted supervision, development, approved Cold-Val studies and final label access. All candidates retain their ranking positions; final predictions are locked before authorized labels are read.

original ESCI paper's $1/0.1/0.01/0$ gain convention [1, 14]. Warm, Cold-Test, and Overall have distinct masks, eligible queries, and IDCGs, so Overall NDCG is not an average of the subgroup NDCGs.

For catalog retrieval, known-positive recall is

$$\mathrm{KPR}_{q,S}@K = \frac{|\{p_1, \ldots, p_K\} \cap \mathcal{P}_q^S|}{|\mathcal{P}_q^S|}. \tag{9}$$

Hit@50 indicates at least one such positive; judgement coverage measures the fraction of returned positions with an observed judgment in the target scope. Unknown products never receive an I label. Full-catalog NDCG is not computed because judgments are incomplete. Reranking the same Top100 cannot change KP-R@100, although it can change smaller-cutoff recall.

Fixed-pool and catalog metric averages are query macros; separately reported unweighted fold means are labeled as such. Final paired comparisons use 2,000 saved query-level bootstrap resamples, seed 20261015, and percentile 95% intervals. Relative-gain intervals resample the ratio of means, rather than averaging per-query percentage gains. Development experiments use their separately recorded budgets, including 1,000 resamples for Week6 ranking and 5,000 for the image ablation. Intervals are not multiplicity-adjusted, and do not include retraining-seed uncertainty. Alignment intervals resample product queries against a fixed gallery. The final effective populations are 1,337 Warm, 1,012 Cold-Test, and 1,339 Overall queries, with 17,437, 2,208, and 19,645 known E/S pairs respectively.

4.4 Training, Selection, and Compute

Week2 uses squared-error XGBoost with depth 6, learning rate 0.05, minimum child weight 10, and L2 penalty 10. Week5 LambdaMART uses 31 leaves, learning rate 0.05, minimum leaf count 20, and NDCG@10-oriented gain settings. The ListNet MLP uses Adam with learning rate 10^{-3}, weight decay 10^{-5}, batches of 64 query lists, at most 30 epochs, and early stopping. The projection study uses five epochs, batch size 256, and AdamW with learning rate 10^{-4}. Configurations, selected checkpoints, and feature/scaler identities are retained in the appendix and source evidence map.

The final experiment's cached CPU timing, the later A100 and T4 engineering sessions, and local CPU online serving are different measurements. Same-session comparisons use identical query sets, thread settings, and cache rules. In the T4 Colab session, the optimized core remains CPU-based FAISS and NumPy computation; the presence of a GPU does not make the measured core a GPU pipeline. The local live demo and its repeated evaluation use an Intel Core i5-1038NG7 CPU, four physical/eight logical cores, and 16 GiB RAM. The repeated study fixes four numerical threads,

declares A/B/C timing boundaries and independently conditions each product-cache workload. No cross-machine speedup is inferred. The released final verification checks prediction locks, model dependencies, and saved metric arithmetic; the report's tables are generated from these archived records, without new training or relevance evaluation.

5 Experimental Results and Analysis

5.1 Final Ranking in the Fixed Judged Pool

Table 2 presents the one-time final evaluation. The selected ListNet has the highest Warm NDCG@10 point estimate, 0.7163, compared with 0.6740 for the original Week2 XGBoost and 0.6660 for BM25. LambdaMART reaches 0.7131. On Cold-Test, however, LambdaMART has the highest NDCG point estimate, 0.3320, while ListNet reaches 0.3247. The formal ranker remains the ListNet chosen during development; the final comparison is not used to switch models.

The paired contrasts in Table 3 and Figure 3 distinguish measured improvements from the original project targets. Week2 XGBoost improves Warm NDCG by 1.20%, below the required 5%. Week5 LambdaMART improves by 5.80% over that XGBoost, below 12%. ListNet's corresponding improvement is 6.28%, but ListNet does not substitute for a requirement that names LambdaMART. The final ListNet's Cold-Test improvement is 1.69%, below 15%, with an absolute-difference interval of $[-0.0098, 0.0208]$. All three ranking targets are unmet. The 164,900-product exact index meets the original 100K size requirement. Final-run cached FAISS search averages 65.24 ms with P95 69.14 ms, however, so the original sub-50 ms index-search target is unmet in that environment. The 0.0033 Warm advantage of ListNet over LambdaMART also has an interval spanning zero, so the final test does not establish stable superiority between those two rankers.

Table 2: One-time Final Test ranking in the fixed judged candidate pool.

Frozen model	NDCG@10	MAP@10	MRR
Warm ($n = 1,337$; E/S pairs = 17,437)			
BM25	0.6660 [0.6561, 0.6756]	0.5785 [0.5655, 0.5914]	0.8148 [0.8000, 0.8296]
Week2 XGBoost	0.6740 [0.6637, 0.6835]	0.5877 [0.5746, 0.6001]	0.8301 [0.8159, 0.8449]
LambdaMART (28)	0.7131 [0.7037, 0.7222]	0.6228 [0.6111, 0.6347]	0.8520 [0.8377, 0.8655]
ListNet (28)	**0.7163** [0.7070, 0.7255]	**0.6278** [0.6154, 0.6398]	**0.8523** [0.8384, 0.8658]
Cold-Test ($n = 1,012$; E/S pairs = 2,208)			
BM25	0.3090 [0.2932, 0.3246]	0.1885 [0.1752, 0.2013]	0.2895 [0.2716, 0.3069]
Week2 XGBoost	0.3193 [0.3032, 0.3350]	0.1982 [0.1845, 0.2118]	0.2979 [0.2794, 0.3161]
LambdaMART (28)	**0.3320** [0.3146, 0.3483]	**0.2105** [0.1954, 0.2249]	**0.3090** [0.2898, 0.3276]
ListNet (28)	0.3247 [0.3081, 0.3402]	0.2073 [0.1923, 0.2214]	0.3036 [0.2854, 0.3211]
Overall ($n = 1,339$; E/S pairs = 19,645)			
BM25	0.7234 [0.7140, 0.7325]	0.6828 [0.6692, 0.6961]	0.8660 [0.8521, 0.8791]
Week2 XGBoost	0.7339 [0.7246, 0.7431]	0.6972 [0.6841, 0.7103]	0.8850 [0.8722, 0.8977]
LambdaMART (28)	0.7758 [0.7671, 0.7839]	0.7355 [0.7233, 0.7470]	**0.9118** [0.9000, 0.9226]
ListNet (28)	**0.7776** [0.7689, 0.7857]	**0.7389** [0.7265, 0.7502]	0.9095 [0.8975, 0.9207]

Notes. Cells show query macro means and saved 95% query-bootstrap intervals. All competitor positions remain; only target-scope gains are masked. MRR is untruncated. Warm, Cold-Test and Overall have distinct eligibility/IDCG; Overall is not an average of the subgroup scores. Bold marks the highest point estimate within a scope and metric, without a significance claim.

Table 3: Original NDCG targets and paired Final Test contrasts.

Contrast	Scope	Target	Relative gain	Δ NDCG	95% CI	Outcome
Week2 XGBoost / BM25	Warm	5%	1.20%	+0.0080	[0.0014, 0.0149]	Unmet
LambdaMART / Week2 XGBoost	Warm	12%	5.80%	+0.0391	[0.0311, 0.0471]	Unmet
Final ListNet / Week2 XGBoost	Cold-Test	15%	1.69%	+0.0054	[-0.0098, 0.0208]	Unmet
Additional paired Warm Test contrasts (not substitutes for the named gates)						
ListNet / Week2 XGBoost	Warm	—	6.28%	+0.0423	[0.0342, 0.0505]	Descriptive
ListNet / LambdaMART	Warm	—	0.46%	+0.0033	[-0.0007, 0.0073]	Descriptive

Notes. Relative gains use the named baseline on the same Test scope. Intervals are for absolute paired NDCG differences (2,000 saved resamples, seed 20261015). The ListNet–LambdaMART interval crosses zero. The Week5 requirement names LambdaMART; ListNet cannot replace it.

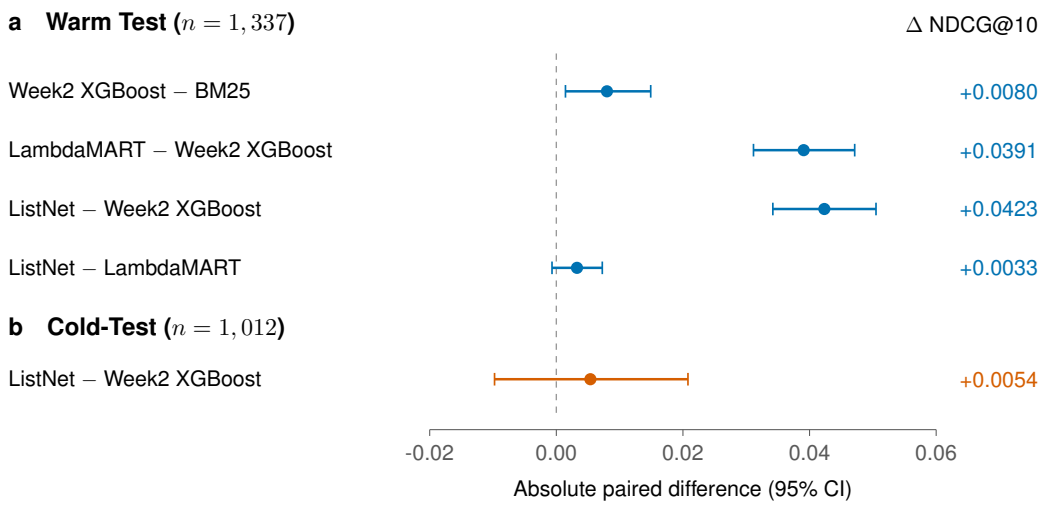


Figure 3: Paired ranking effects on the sealed Final Test. Points show absolute NDCG@10 differences and horizontal bars show the saved query-bootstrap 95% intervals. Warm and Cold-Test use their own target masks and eligible queries. The dashed zero line indicates no change; intervals spanning it do not establish a positive effect. No comparison is used to change the frozen system.

These comparisons answer different questions. Week2 tests a learned pointwise model against lexical retrieval scores. Week5's target comparison adds both semantic/visual features and a different objective relative to Week2's 20-feature model; it is not an isolated causal estimate of the ranking loss. The development same-28-feature experiment is the appropriate control for comparing model families. Warm and Overall performance must also be interpreted independently of Cold performance, because their target masks and ideal rankings differ.

5.2 Retrieval, Calibration, and the Auxiliary RRF Flow

5.2.1 Why calibration was necessary

Initial Week4 experiments show that better ranking of a small judged pool need not imply better catalog coverage. The original late fusion reaches Warm Validation NDCG@10 of 0.6861, but its catalog KP-R@100 of 0.5344 is below SBERT's 0.5766. On image-available judged candidates, late fusion's NDCG of 0.7004 exceeds SBERT's 0.6891 and CLIP text's 0.6714. These observations motivate an explicit score-scale and fallback diagnosis rather than a general claim that images are harmful or helpful.

Unlabeled calibration scores provide a concrete explanation for the original availability bias. Across 65,536 Train score pairs, CLIP text cosine on image-available products has mean 0.2256 and standard deviation 0.1250, whereas image cosine has mean 0.0994 and standard deviation 0.0411. Missing-

Table 4: Week4 Warm Validation retrieval trajectory and fixed calibration controls.

Route / configuration	Pool NDCG@10	Catalog KP-R@50	Catalog KP-R@100	No-image Top10
Original Warm Validation evaluation (full judged pool / full catalog)				
BM25	0.6642	**0.4942**	0.5753	7.28%
SBERT text	0.6754	0.4783	**0.5766**	9.49%
CLIP text only	0.6587	0.2316	0.2806	14.68%
CLIP image route	0.6665	0.0234	0.0259	99.96%
CLIP text + image	0.6738	0.0347	0.0446	99.68%
Original late fusion	**0.6861**	0.4385	0.5344	53.96%
Supervised projection	0.6294	0.0262	0.0425	24.03%
Bounded Train-only calibration study, same Warm evaluation scope				
Affine calibrated fusion	**0.6918**	0.5429	0.6366	19.38%
Calibrated + mask-aware fusion	0.6898	**0.5570**	**0.6533**	2.07%

Image-available diagnostic	Pool NDCG@10
Separate image-available candidate subset, shared by all routes below	
SBERT text	0.6891
CLIP text only	0.6714
CLIP text + image	0.6971
Original late fusion	**0.7004**

Notes. The upper panels use 1,343 E/S-eligible Validation queries; pool NDCG preserves original competitor positions and targets Warm products. Catalog KP-R uses 16,839 known Warm E/S pairs, with unknown products unlabeled. No-image Top10 is the product-slot share over all 1,344 Validation queries, not the E/S-eligible denominator. Original late fusion selected $\alpha = \beta = 0.5$ by pool NDCG; the bounded calibration study selected mask-aware retrieval by catalog KP-R@50, then KP-R@100. The lower panel is a separate shared image-available candidate subset (1,331 E/S-eligible queries); its scores must not be compared as if the candidate pool were unchanged. The projection row uses the original selected supervised projection, which was not adopted. Bold marks the highest metric point estimate separately within the original, calibration and image-available panels, without a significance claim; the no-image slot share is a diagnostic, not a best-value criterion.

image text cosine has mean 0.2946 and standard deviation 0.1612. A direct text fallback thus changes both location and spread. The calibrated mask-aware rule brings Warm Validation KP-R@50 to 0.5570 and KP-R@100 to 0.6533; calibration without the mask-aware branch reaches 0.5429 at 50. This is a development-selected result with fixed Train-only calibration parameters. Missing-image recall remains a recorded tradeoff, and no further fallback tuning is performed.

5.2.2 Final catalog results

Table 5 separates raw retrieval from reranked flows. Raw calibrated Fusion reaches Warm KP-R@50 of 0.5449, compared with 0.4765 for SBERT: an absolute gain of 0.0684 and a relative gain of 14.35%. The paired absolute interval is $[0.0585, 0.0779]$. This exceeds the original 10% retrieval-improvement target under the registered known-positive proxy, and supports a gain in coverage of known E/S products. It does not establish recall over every relevant catalog product, because most returned products have no judgment. Warm coverage at rank 50 is only 0.1615 for raw Fusion, which makes that distinction practically important.

Formal flow A, Fusion Top100 followed by ListNet, reaches KP-R@50 of 0.5641 Warm, 0.5631 Cold-Test, and 0.5631 Overall. Auxiliary flow B replaces only retrieval with RRF60 and reaches 0.5967, 0.5862, and 0.5951. The Cold-Test difference is $+0.0230$ with interval $[0.0088, 0.0371]$. Smaller-cutoff recall can improve through both candidate coverage and reranking, while KP-R@100 reflects candidate coverage alone. The BM25-to-Week2-XGBoost flow is also retained as a registered baseline; its reranking does not improve all catalog recall measures. A ranker trained on judged lists need not preserve catalog recall at every cutoff.

Table 5: Final Test catalog retrieval and registered reranking flows.

Route / flow	KP-R@10	KP-R@50	KP-R@100	Hit@50	Cov.@50
Warm ($n = 1,337$)					
BM25	0.2508	0.4989	0.5818	0.8945	0.1526
SBERT	0.2282	0.4765	0.5692	0.8901	0.1407
Calibrated Fusion	0.2693	0.5449	0.6488	0.9080	0.1615
RRF60	**0.2823**	**0.5678**	**0.6778**	**0.9349**	**0.1693**
D1: BM25 → Week2 XGB	0.2349	0.4718	0.5818	0.8788	0.1428
A: Fusion → ListNet	0.2964	0.5641	0.6488	0.9252	0.1665
B: RRF60 → ListNet	**0.3023**	**0.5967**	**0.6778**	**0.9521**	**0.1758**
Cold-Test ($n = 1,012$)					
BM25	0.2357	0.4839	0.5862	0.6275	0.0245
SBERT	0.2268	0.4680	0.5716	0.6057	0.0226
Calibrated Fusion	0.2465	0.5461	0.6456	0.6828	0.0261
RRF60	**0.2677**	**0.5698**	**0.6746**	**0.7105**	**0.0275**
D1: BM25 → Week2 XGB	0.2229	0.4717	0.5862	0.6245	0.0234
A: Fusion → ListNet	0.2781	0.5631	0.6456	0.7045	0.0270
B: RRF60 → ListNet	**0.2823**	**0.5862**	**0.6746**	**0.7233**	**0.0280**
Overall ($n = 1,339$)					
BM25	0.2498	0.4977	0.5820	0.9014	0.1717
SBERT	0.2275	0.4750	0.5686	0.9029	0.1582
Calibrated Fusion	0.2679	0.5449	0.6472	0.9149	0.1818
RRF60	**0.2806**	**0.5679**	**0.6772**	**0.9410**	**0.1907**
D1: BM25 → Week2 XGB	0.2348	0.4715	0.5820	0.8895	0.1610
A: Fusion → ListNet	0.2962	0.5631	0.6472	0.9343	0.1874
B: RRF60 → ListNet	**0.3025**	**0.5951**	**0.6772**	**0.9552**	**0.1974**

Notes. Catalog size 164,900; each route and reranked flow returns Top100. First four rows of each scope are raw retrieval; D1/A/B are reranked flows. D1 uses the original Week2 XGBoost; A and B share the original frozen 28-feature ListNet. KP-R is known-positive recall over judged E/S only; coverage is the judged target fraction. Unjudged products stay unknown. Bold is scoped separately to raw routes and reranked flows. A remains formal; B is auxiliary.

5.2.3 RRF chronology and adoption boundary

Table 6 retains the development sequence rather than reporting only the favorable final auxiliary comparison. Internal grouped experiments and Warm Validation show positive RRF gains. The approved Cold-Val safety comparison has a smaller $+0.0104$ KP-R@50 difference and an interval spanning zero. It therefore fails the prespecified Cold promotion rule. RRF is retained as a preregistered auxiliary flow, not the formal system. The subsequent positive final auxiliary result does not reopen model selection.

Subgroup and cost evidence qualifies this alternative. Cold-Val raw Top100 gains 120 known Cold E/S products while losing 98; individual queries can therefore regress despite an average coverage gain. Missing-image Cold-Val queries benefit more strongly, but there are only 84 eligible queries, and image-available Recall@10 has no clear improvement. The extra routes add approximately 377 ms in the Warm session and 508 ms in the Cold safety session, with about 600 MiB additional memory. These costs were measured before the later formal-core engineering optimizations and should not be extrapolated to an unmeasured optimized RRF service. Appendix E reports the final image, missing-image, and content-unique subsets without treating them as new selection sets.

5.3 Feature Ablations and the Role of Images

The controlled Week5 comparison uses the same 28-feature training view and judged-pool protocol. On Warm Validation, ListNet reaches NDCG@10 of 0.7047, compared with 0.6995 for both tree models at the displayed precision. This selected the frozen ListNet. Table 7 reports deletion-and-retraining ablations, preserving all candidate competitors and the original target mask.

Table 6: RRF60 chronology: same-budget recall and added cached-CPU cost.

Protocol / target	A KP-R@50	B KP-R@50	Δ(B–A)	Paired 95% CI
Internal Warm / OOF	0.5880	0.6147	+0.0267	[0.0237, 0.0298]
Internal pseudo-Cold / clean OOF	0.5926	0.6150	+0.0224	[0.0172, 0.0276]
Warm Validation	0.5794	0.6040	+0.0246	[0.0181, 0.0307]
Cold-Val	0.5903	0.6006	+0.0104	[-0.0023, 0.0236]
Final Test / Warm	0.5641	0.5967	+0.0327	[0.0261, 0.0391]
Final Test / Cold-Test	0.5631	0.5862	+0.0230	[0.0088, 0.0371]
Final Test / Overall	0.5631	0.5951	+0.0320	[0.0259, 0.0387]

Cached CPU timing boundary	A mean (ms)	B mean (ms)	Paired increase (ms)
Internal raw retrieval	66.4	93.6	+27.2
Warm Validation / full flow	2315.0	2692.1	+377.2
Cold-Val / full flow	2796.1	3304.3	+508.2
Final Test / full flow	2043.7	2458.7	+415.0

Notes. A is calibrated Fusion and B is RRF60, followed by the same original-feature ListNet appropriate to each protocol. Internal rows use query macro OOF estimates; clean pseudo-Cold excludes held-out product supervision. External rows use the frozen original ListNet. Each of BM25/SBERT/Fusion supplies Top100; RRF merges to Top100 with $k = 60$. Internal timing covers retrieval only; external timing covers the complete cached CPU flow. Both exclude online encoders/HTTP; no cross-machine speedup is inferred. B remains auxiliary despite the favorable final point estimates.

Table 7: Warm Validation feature-group retraining ablations.

Feature set	XGBoost (28)	LambdaMART	ListNet
Full (28)	0.6995	0.6995	**0.7047**
Remove Lexical	0.6889	0.6912	**0.6952**
Remove Semantic	0.6948	0.6962	**0.6968**
Remove Image	0.6882	0.6921	**0.6940**
Remove Metadata	0.6947	0.7005	**0.7016**
Paired ListNet image comparison: full minus no-Image			
Δ NDCG@10	+0.010782	95% CI	[0.004866, 0.016676]

Notes. NDCG@10 over 1,343 E/S-eligible queries, retaining all competitors. Groups overlap through shared features; no-Image removes image similarity, availability and image-dependent fusion while retaining pure text fusion. The saved paired ListNet interval uses 5,000 development-query resamples; selection on the same Validation set limits inference. Bold compares model families only within a row. These ablations were not rerun on Final Test.

Removing the Image group lowers ListNet from 0.7047 to 0.6940, an absolute difference of 0.010782 with paired interval $[0.004866, 0.016676]$. This is evidence for the implemented image-dependent feature group in this development setting. The removal includes image similarity, the availability mask, and image-dependent fusion; pure text fusion is retained. It does not isolate pixels from all related score-processing effects. Groups also overlap through the calibrated fusion feature, so their deltas cannot be summed as independent contributions. Metadata removal improves LambdaMART slightly, while reducing ListNet; a feature's utility depends on its model and interactions. No new Final Test image ablation is run.

5.4 Cold Generalization and Projection Adaptation

5.4.1 Final product-holdout ranking

Cold-Test evaluates products excluded from supervised model training, with their content still available at inference. ListNet's fixed-pool NDCG@10 of 0.3247 exceeds Week2 XGBoost's 0.3193 by only 0.0054, with uncertainty spanning zero. This is not evidence that the system has achieved the 15% Cold ranking target. The content-unique Cold subset retains 982 eligible queries and

2,085 E/S pairs; ListNet reaches 0.3209 versus XGBoost's 0.3182, again a small point difference. Content-overlap removal therefore does not reveal a large hidden gain.

Image subgroups differ substantially. On 979 image-available Cold queries, ListNet NDCG is 0.3145 versus XGBoost's 0.3179. On 86 missing-image Cold queries, it is 0.3902 versus 0.2792. These are target-specific metrics with different eligible populations, and the small missing-image subset is not a basis for a universal fallback conclusion. Train-Seen diagnostics additionally show that many Warm target products were never present in actual supervised training; a nominal Warm split should not be equated with observed interaction history.

5.4.2 Why the projection alternative was not retained

The projection experiment trains on 88,063 permitted product pairs and evaluates against a fixed 16,853-product alignment gallery. Although a later adapted checkpoint is selected among the adapted epochs, it remains below the original Frozen CLIP baseline. Mean bidirectional Recall@10 decreases from 0.834896 to 0.828072; both directional Recall@1 measures decline. Table 8 connects the alignment and downstream evaluations.

Cold-Val judged-pool NDCG rises from 0.3292 to 0.3340, but its paired interval spans zero. Warm NDCG decreases from 0.7047 to 0.7036, and Cold catalog KP-R@50 falls from 0.5903 to 0.5741, with a negative paired interval. The adapter is therefore not adopted. Better optimization of a product-pair training loss would not, by itself, establish query-ranking generalization. The unchanged ListNet may also be sensitive to altered feature relationships after projection, but this study does not factorially separate that mechanism from changes in retrieval or representation quality. The observed alignment decline is already sufficient to reject a simple claim of successful adaptation.

5.5 Controlled Optimization and Negative Results

The final targets motivate bounded follow-up experiments rather than unrestricted tuning. Table 9 summarizes every principal ranking direction. Original models and results remain archived, and all new ranking choices are made inside Train. P1 uses a fixed grouped 80/20 split; P2 and the final targeted experiments use matched three-fold grouped comparisons, with a product holdout when evaluating pseudo-Cold generalization.

Weights and regularization. The Week2 Warm training view is positive-heavy: 80,216 of 101,984 pairs are E/S, and 2,261 queries have no Warm C/I negatives. This suggests testing query-balanced

Table 8: Frozen versus projection-adapted development results.

Encoder heads	T→I R@1	I→T R@1	T→I R@10	I→T R@10
Warm paired-product alignment (fixed gallery of 16,853 products)				
Frozen	**0.4395**	**0.4565**	**0.8276**	**0.8422**
Adapted	0.4225	0.4457	0.8228	0.8333

Judged-pool scope	n	Frozen	Adapted	Δ NDCG	95% CI
Warm Validation	1,343	**0.7047**	0.7036	-0.0012	[-0.0058, 0.0036]
Cold-Val	996	0.3292	**0.3340**	+0.0048	[-0.0040, 0.0127]
Overall Validation	1,344	**0.7654**	0.7650	-0.0004	[-0.0046, 0.0036]

Reranked catalog KP-R@50	Frozen	Adapted
Warm Validation	**0.5794**	0.5663
Cold-Val	**0.5903**	0.5741
Overall Validation	**0.5793**	0.5656

Notes. Mean bidirectional alignment R@10 decreases from 0.834896 to 0.828072. Pool cells are NDCG@10; deltas are adapted minus frozen. Cold-Val catalog KP-R@50 changes by -0.016146 (95% CI [-0.031109, -0.001424]). The Warm non-degradation rule fails; the frozen system is retained. This is development evidence, with an unchanged frozen ranker. Bold marks the higher point estimate within each matched metric/scope comparison, without a significance claim.

Table 9: Controlled ranking extensions and their adoption outcomes.

Candidate vs matched control	Protocol	Δ NDCG	Paired 95% CI	Decision
XGB: NDCG stopping	Internal 80/20	+0.00102	[-0.00309, 0.00501]	Not promoted
XGB: query weights	Internal 80/20	+0.00296	[-0.00262, 0.00852]	Not promoted
XGB: hard negatives	Internal 80/20	-0.00047	[-0.00576, 0.00461]	Not promoted
LambdaMART: L2=5	Internal 80/20	-0.00007	[-0.00295, 0.00284]	Not promoted
LambdaMART: 15 leaves	Internal 80/20	+0.00061	[-0.00220, 0.00355]	Not promoted
Constraints / XGB	Grouped 3-fold OOF	+0.00176	[0.00017, 0.00335]	Not promoted
Constraints / LambdaMART	Grouped 3-fold OOF	+0.00154	[0.00012, 0.00298]	Not promoted
Constraints / ListNet	Grouped 3-fold OOF	+0.00136	[0.00040, 0.00240]	Not promoted
Gain-softmax / ListNet	Grouped 3-fold OOF	-0.00062	[-0.00156, 0.00031]	Not promoted
Position-aware loss	Grouped 3-fold OOF	+0.00050	[-0.00107, 0.00198]	Not promoted
Field features / LambdaMART	Grouped 3-fold OOF	+0.00039	[-0.00101, 0.00178]	Not promoted
Field features / ListNet	Grouped 3-fold OOF	-0.00057	[-0.00168, 0.00043]	Not promoted
Position loss / pseudo-Cold	OOF pseudo-Cold	-0.00301	[-0.00578, -0.00010]	Not promoted
Projection heads	Cold-Val	+0.00481	[-0.00403, 0.01266]	Warm safety fails

Notes. All deltas use query macro means; they are not fold-mean deltas. P1 uses a fixed grouped internal split and its selection-affected interval. P2/targeted ranking rows use three grouped out-of-fold (OOF) comparisons and require an absolute gain of at least 0.002, a positive interval lower bound, and safety. Projection is a separate Cold-Val experiment requiring Cold improvement and Warm non-degradation. OOF query bootstrap does not fully capture fold clustering or training-seed variation. P2C and targeted D are recall experiments and appear separately in Table 6.

weighting and modest upweighting of already judged lexical hard negatives. Query-balanced XGBoost has a $+0.00296$ internal point difference, but its interval spans zero. NDCG-based stopping, hard-negative weighting, and the two tree-regularization settings also lack stable improvement. These results do not justify larger searches. The imbalance diagnosis identifies a limitation of the available supervision, but does not demonstrate that a particular weighting scheme solves it.

Constraint and field features. Five general features address model identifiers, numbers, product type, accessory relations, and negation. The three model families all have small positive out-of-fold differences, and their paired intervals exclude zero. However, their absolute differences—0.00176, 0.00154, and 0.00136—are below the prespecified 0.002 promotion threshold. The distinction between statistical evidence for a small effect and the minimum practical effect is retained. A later field-level matching study does not provide a stronger result: LambdaMART changes by $+0.00039$ and ListNet by -0.00057, with both intervals spanning zero. These are evaluated features, not features silently added to the final 28-column contract.

Ranking losses. Gain-aware ListNet changes NDCG by -0.00062, so closer resemblance between a target distribution and relevance gain does not translate into better ranking. The position-aware pair objective changes Warm NDCG by $+0.00050$ with an interval spanning zero. On pseudo-Cold products it decreases NDCG by 0.00301, with interval $[-0.00578, -0.00010]$. That experiment excludes all supervision for 18,758 held-out Warm product IDs and retains the full competitor lists. It gives direct negative evidence for this specific loss and training budget, rather than evidence that product-holdout ranking can be improved through loss replacement alone.

The raw Week4 projection-fusion attempt, the Week6 contrastive heads, P1, P2-A/B, and targeted A/B/C all remain negative or unpromoted results. Among these controlled follow-ups, the retrieval di-

rection with recurring positive evidence is RRF60, whose Cold-Val safety uncertainty and deployment cost are reported separately. None of these outcomes changes the frozen final system.

5.6 Error Cases and Historical Evaluation

5.6.1 Hard-case categories in the saved examples

The archived selectors preserve five improvements and five degradations for several model contrasts. For "cartera de guchi", ListNet moves an E-labeled Gucci wallet from below the saved top ten to first place, while an I-labeled Gucci perfume falls from first to below the top ten; the Warm NDCG difference is $+0.8401$. This is visible product-type discrimination under a noisy query, not proof that the model explicitly corrects spelling. For "modern ceiling fan without light", two I-labeled with-light fans rise, and the difference is -0.5197. Yet the previous first product also says "with Light" and is labeled E. The ranking decline is real relative to the released judgments, while a purely title-based negation explanation is incomplete.

The same ambiguity occurs for "food freeze dryer machine not dehydrator": promoted dehydrators are labeled I, but some previous dehydrator titles are labeled E. For the Cold query "stand mixer", an E-labeled mixer rises while a C-labeled cover falls; another cover is itself labeled E. For the auxiliary catalog query "truly's", a matching deodorant rises to first place, but sparse saved top-ten judgments cannot identify all Cold positives responsible for a Recall@50 change. These examples motivate attention to product type, exclusions, and label ambiguity without inventing new labels. Appendix H preserves all ten main ListNet/Week2 extreme examples. Extreme-case selection does not estimate failure prevalence.

5.6.2 Historical metadata pipeline

Table 10 separates the older weak/silver evaluation from the current human-judged benchmark. In the later historical fixed-100K study, Week2 XGBoost improves NDCG by 19.00% over BM25; multimodal retrieval improves Recall@50 by 24.01% over raw SBERT; the BGE cross-encoder prior plus residual LambdaMART improves final NDCG by 10.55% over that Week2 baseline. A separate prospective 200-query Week6 evaluation gives a 14.02% Cold NDCG improvement for the expanded pretrained text-reranker ensemble with content-gated compensation. Its Cold eligibility is 95 queries, not the current ESCI Cold population.

Matched historical controls are important. The strong text retriever and image-off version exceed the selected historical multimodal Recall@50, so the 24.01% headline does not isolate a positive image effect. The historical CE-only point estimate also exceeds the CE-plus-residual result, so the combined gain is not evidence of an incremental residual benefit. An earlier 48K weak-label report quotes a 21.15% Week2 improvement, but its complete primary result package is unavailable; it is retained as report-level history rather than substituted for the verified later 19.00% result.

The grade-stratified full-text human review contains 100 pairs, 99 assessable. Exact-grade agreement is 50/99 (50.51%); binary E/S agreement is 70/99 (70.71%). Earlier title-only inspection used less content than the model judge, and the review sequence includes subsequent clarification. This limited audit supports concern about weak-label quality, but it is not an independently labeled complete ranking pool. Larger historical headline gains and smaller ESCI gains therefore cannot isolate architecture or label-source effects. Dataset, queries, relevance policy, candidate pools, and Cold definitions all change. A same-architecture metadata reconstruction was implemented but not run as a new complete training/evaluation study; no such controlled performance claim is made here.

5.7 Efficiency, Online Search, and Concurrency

5.7.1 Equivalent computation and matched comparisons

The original cached flow averages 2,043.7 ms, including approximately 1,861 ms for feature construction. Profiling identifies repeated full-catalog Pandas indexing and field conversion. Prepared28 and aligned candidate-key arrays remove this overhead without changing Top100, any feature formula, the scaler or ListNet. Historical A0/A1/A2 measurements are retained in Table 11 and Figure 4: all variants in that comparison ran in the same T4-equipped Colab session, with the core itself executing on CPU. In that session, A2 averages 94.93 ms with P95 113.35 ms; its 21.97-fold A0/A2 speedup is not calculated across machines.

Table 10: Historical metadata/silver-label outcomes and matched controls.

Historical comparison	Metric	Baseline	System	Rel. gain	Gate
Fixed100K Week2 / BM25	NDCG@10	0.5379	0.6402	19.00%	Met
Fixed100K Week4 / raw SBERT	Recall@50	0.4941	0.6128	24.01%	Met
Fixed100K Week5 / Week2	NDCG@10	0.6402	0.7077	10.55%	Unmet
200-query Week6 / Week2	Cold NDCG@10	0.4804	0.5478	14.02%	Unmet

Matched control	Metric	Score
Matched fixed100K Test controls (same protocol within each block)		
Week4 multimodal	Recall@50	0.6128
Week4 strong text	Recall@50	0.6184
Week4 same-model image off	Recall@50	**0.6300**
Week5 CE + residual	NDCG@10	0.7077
Week5 CE only	NDCG@10	**0.7090**
Separate 200-query confirmation: original positions, Cold target mask		
BGE ensemble without compensation	Cold NDCG@10	0.4975
BGE ensemble + compensation	Cold NDCG@10	0.5475
Expanded ensemble + compensation	Cold NDCG@10	**0.5478**

Notes. These synthetic-query, text-only silver-judgment experiments are separate from ESCI human-judged evaluation. Historical Recall@50 uses common-pool silver grade ≥ 2 positives as its denominator. The fixed100K final ranking comparisons use 100 Test queries; Week6 confirmation uses 200 queries with 95 Cold-eligible queries. Week4 is an expanded BGE/projection/CLIP system, and Week5 is CE + learned residual. Hybrid minus CE-only NDCG is -0.001364 (95% CI [-0.011494, 0.009002]). The earlier 48K +21.15% claim is report-level evidence and is excluded from this numerical comparison. Bold marks the highest point estimate separately within each matched control block and metric, without a significance claim.

Table 11: Historical matched-session cached-core comparison; separate from the local repeated evaluation below.

Implementation	Mean (ms)	P50 (ms)	P95 (ms)	$<$ 200 ms
A0: Original core	2085.944	2078.778	2165.348	0%
A1: Prepared28	226.807	209.103	306.661	34%
A2: Prepared28 + candidate arrays	**94.927**	**94.993**	**113.355**	**100%**

Notes. 100 distinct unlabeled queries, three correlated repeats per implementation (300 records), in the same T4-allocated session using CPU core computation. Complete cached-core latency excludes query encoding and HTTP. Candidates, raw/scaled features, scores and full ranks are identical in all 300 paired triples. A0/A2 mean ratio = 21.974×. Bold marks the lowest recorded latency and highest under-threshold fraction in this same-session comparison; it is descriptive, not a significance claim. This is not a cross-hardware comparison.

5.7.2 Repeated local A/B/C measurements

The subsequent evaluation uses one Intel Core i5-1038NG7 Mac, 16 GiB RAM, macOS 14.8.7 and four numerical threads. Each repeat starts a new API and encoder worker with one A2 core. A starts with cached query vectors and includes Top100 retrieval, candidate construction, all 28 features, scaling, ListNet and sorting. B includes live SBERT/CLIP encoding and product-field construction. C measures the client's complete HTTP body. These boundaries are measured independently: their quantiles are neither added nor interpreted by subtracting C minus B. All B/C requests execute the encoders; no answer or query-vector cache is introduced.

Before a workload-conditioned scene, the product LRU is cleared and the fixed query list runs twice. Both passes record their duration, hits, misses and cache size. Models remain loaded. The Hot64 and Broad1000 workloads are conditioned independently, and warming does not imply that the larger Broad working set fits in the 8,192-product cache. A separate paired 200-query A0/A2 comparison and an empty-cache fill scenario start from an empty product LRU. Detailed steps are provided in Appendix G.3.

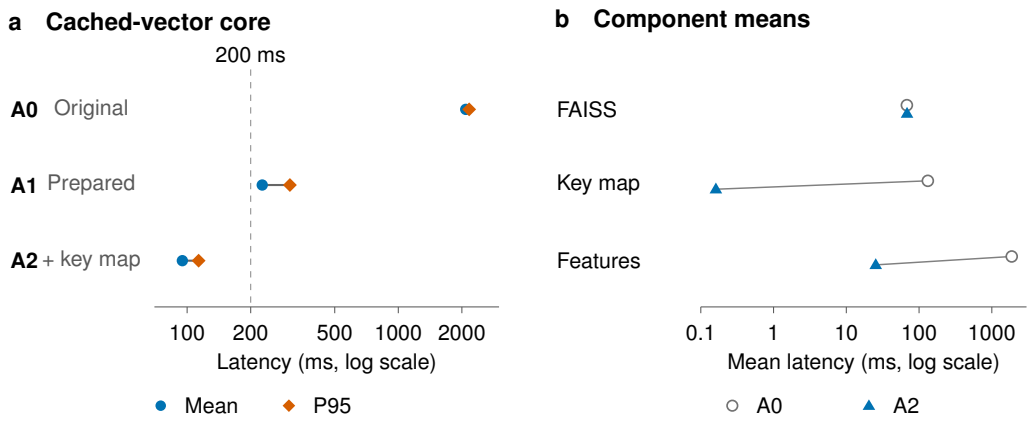


Figure 4: Historical A0/A1/A2 core measurements from one matched Colab session. A GPU-equipped host does not make FAISS/NumPy CPU work GPU inference. These values are not pooled with local CPU results.

Four repeats complete A/B/C, giving 4,000 observations per boundary (Table 12). Mean cached-core latency is 102.16 ms, compared with 164.03 ms for B and 184.52 ms for C. Pooled P95 is 204.99, 337.10 and 370.08 ms respectively. Thus, mean Top100 retrieval and reranking meet 200 ms, but stable P95 below 200 ms is not demonstrated. Figure 5 and Table 13 retain the slower repeats. Under the separate empty-cache paired condition, mean A0/A2 latency is 2,756.88/114.06 ms, a 24.17-fold ratio; all 800 paired complete-array comparisons are exact. That ratio does not mix the empty-cache and conditioned measurements.

5.7.3 Burst and sustained concurrency

Burst64 releases 64 distinct queries through a common gate, drains each wave, waits two seconds, and repeats for 20 waves. Sustained64 runs 64 closed-loop clients with at most one unfinished HTTP request per client, zero think time, a planned 600-second issuing window and final draining. Hot64 and Broad1000 are separate experiments. The queue, single compute slot and original 9.5/10-second queue/request deadlines remain fixed; there are no automatic retries. Thirty-second records preserve offered load, errors, cache, queue and resources. Every failed request stays in the accounting and all-status latency distribution.

Table 12: Repeated local CPU performance with distinct timing boundaries.

Timing boundary	Repeats	N	Mean ± SD (ms)	P95	P99	< 200 ms
A: cached vectors	4	4,000	102.16 ± 48.44	204.99	292.26	94.15%
B: live encoders	4	4,000	164.03 ± 79.81	337.10	401.18	75.00%
C: complete HTTP	4	4,000	184.52 ± 71.15	370.08	593.74	68.17%

Notes. Mean and sample SD are over four repeat means. Pooled percentiles use the corresponding 4,000 raw observations. Each boundary uses the same local CPU, workload and independent two-pass product-cache conditioning; A/B/C are not added or subtracted.

Table 13: Per-repeat local A/B/C means and P95.

Repeat	A Mean / P95 (ms)	B Mean / P95 (ms)	C Mean / P95 (ms)
1	77.83 / 98.08	135.66 / 193.08	269.53 / 550.95
2	174.65 / 265.75	282.88 / 392.42	216.68 / 282.60
3	82.24 / 105.43	125.21 / 152.95	131.59 / 161.01
4	73.93 / 90.27	112.36 / 135.60	120.29 / 144.49

Notes. The slower repeats remain in the summary. These four repeats provide descriptive repeated measurements, not a completed five-repeat certification or new independent query samples.

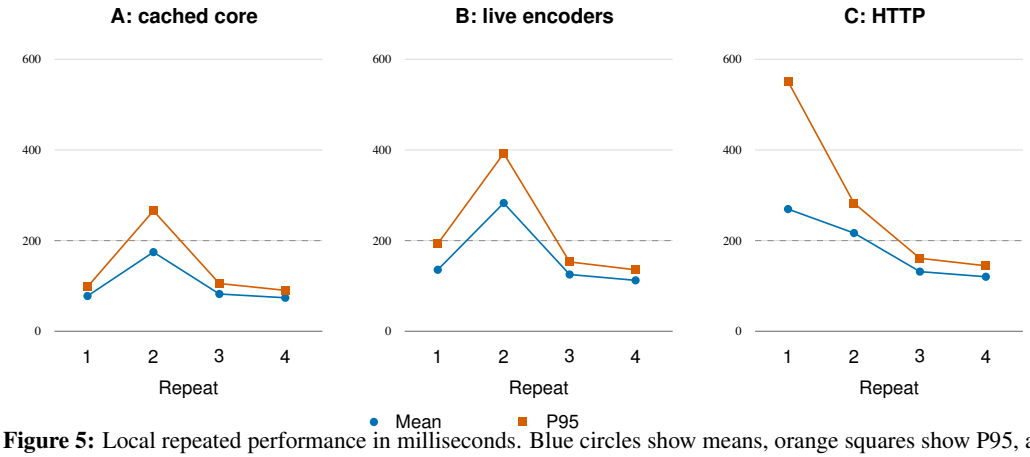


Figure 5: Local repeated performance in milliseconds. Blue circles show means, orange squares show P95, and dashed lines mark 200 ms. Each panel has its own timing boundary but the same local hardware session. Four repeats of fixed inputs are not 4,000 independent query identities.

Table 14 gives each observed repeat. All four Hot burst and sustained loads complete their planned windows, but Hot burst repeat 2 has 170 timeouts, and the first three Hot sustained repeats have 14, 42 and 52 timeouts. One successful Hot response is coalesced; it remains in HTTP success counts but is not additional independent inference. Broad burst has 340 and 64 timeouts in repeats 1–2 and none in repeats 3–4. Broad sustained repeats 1–2 stop issuing at 470.97 and 294.11 seconds because timed-out keys remain quarantined and available query supply is exhausted. Repeats 3–4 complete 600 seconds without HTTP failures. The no-request remainder of an early-terminated window is not treated as faster service.

Successful 64-client requests spend approximately 6.18–9.11 seconds in the queue across these scenes, while their mean internal service computation is approximately 99–197 ms. A single compute slot explains why responsive serial use does not imply short saturated HTTP latency. This is a descriptive mechanism, not a causal comparison of alternative dispatchers. Only complete equal-load

Table 14: Local Hot/Broad concurrency, including failures and incomplete planned loads.

Workload	Rep.	Sent	200 / 504	Mean / P95 (s)	Window (s)	Full load
Hot burst	1	1,280	1,280 / 0	3.48 / 6.35	173.50	Yes
Hot burst	2	1,280	1,110 / 170	5.56 / 10.01	229.71	Yes
Hot burst	3	1,280	1,280 / 0	3.75 / 6.83	183.41	Yes
Hot burst	4	1,280	1,280 / 0	3.36 / 6.13	170.38	Yes
Broad burst	1	1,280	940 / 340	6.69 / 11.16	251.77	Yes
Broad burst	2	1,280	1,216 / 64	5.05 / 10.43	221.99	Yes
Broad burst	3	1,280	1,280 / 0	4.31 / 7.81	205.77	Yes
Broad burst	4	1,280	1,280 / 0	4.18 / 7.60	200.19	Yes
Hot sustained	1	5,947	5,933 / 14	6.50 / 7.04	600.00	Yes
Hot sustained	2	5,671	5,629 / 42	6.80 / 7.68	600.00	Yes
Hot sustained	3	5,489	5,437 / 52	7.03 / 8.39	600.00	Yes
Hot sustained	4	6,123	6,123 / 0	6.30 / 6.59	600.00	Yes
Broad sustained	1	3,433	2,865 / 568	8.85 / 9.80	470.97	No
Broad sustained	2	2,032	1,514 / 518	9.40 / 9.74	294.11	No
Broad sustained	3	4,819	4,819 / 0	8.01 / 8.47	600.00	Yes
Broad sustained	4	5,184	5,184 / 0	7.45 / 7.83	600.00	Yes

Notes. Burst: twenty waves of 64 distinct queries, a common release gate, draining and a two-second gap. Sustained: 64 closed-loop clients, zero think time, planned 600 seconds and draining. All statuses remain in the latency denominator. Broad repeats 1–2 ended early after query-supply exhaustion; their zero-load remainder is not counted as throughput improvement.

Table 15: Repeated concurrency summaries for completed equal-load scenes.

Workload	Complete repeats	Mean ± SD (s)	P95 (s)	P99 (s)
Hot burst	4	4.04 ± 1.03	9.01	10.05
Broad burst	4	5.06 ± 1.15	10.45	11.19
Hot sustained	4	6.66 ± 0.33	7.67	9.50
Broad sustained	2	7.73 ± 0.40	8.36	8.66

Notes. Complete-load aggregates only. All four observed Broad sustained attempts, including both early terminations, are retained in the preceding table. This separation prevents mixing different actual issuing windows while keeping failures visible. Four repeats, or two for complete Broad sustained loads, provide limited evidence about long-term stability.

repeats enter the corresponding aggregate; the two early Broad terminations remain explicitly reported. Independent compute-job throughput is not identifiable from HTTP/ASGI traces alone.

The planned five-repeat attempt stops in repeat 4 during conditioning for the next scene, after a power-observer identity exception. Repeats 1–3 reach all 15 scenes; repeat 4 completes 12 measured scenes and repeat 5 never starts. Completed A/B/C and concurrency measurements are reported with their actual repeat counts, not as a completed five-repeat certification. Across all issued HTTP requests, 67,032 return 200, 1,911 return 504 and 93 return 429; this accounting total is not a pooled performance metric. Successful Top10 predictions and scores match the locked references. Supplemental shutdown evidence records closure of the remaining owned service and encoder without rewriting missing original completion artifacts. No new training or relevance evaluation is performed.

The English frontend uses FastAPI and real catalog titles, IDs, ranks and image URLs (Figure 6). It supports arbitrary natural-language queries using the previously approved encoder-compatibility criterion. API and browser tests remain functional evidence, not extra timing samples. The system is suitable for interactive local demonstration, but these broader repeated measurements do not establish stable error-free 64-client serving. Earlier finite-workload successes are retained separately in the appendix.

6 Discussion

6.1 Annotation Quality and Ambiguous Relevance

Reliable relevance judgments are essential for supervised learning and credible evaluation. In the historical full-text review, exact-grade agreement is only 50.51% across 99 assessable pairs, which motivates more careful annotation rather than accepting model-generated labels as ground truth. The human-judged ESCI benchmark provides a stronger relevance reference. Its higher absolute Warm scores for several baselines do not, by themselves, measure an improvement caused by annotation: queries, products and candidate pools also differ, and BM25 does not learn from relevance labels. Future model-assisted annotation should expose the full available product text, define E/S/C/I consistently, and state how to judge model compatibility, size, accessories, exclusions and missing evidence. A larger annotator, such as a 32B model, is a candidate to evaluate against independently reviewed examples; the present study establishes no reliable model-size threshold. Detailed prompts and human adjudication remain important because capable judges can retain systematic biases [15].

6.2 Week2: Learned Patterns and Product Constraints

XGBoost's positive Warm point estimate is consistent with learning useful ranking patterns from its supplied features, but aggregate relevance is not the same as satisfying every product requirement. The original lexical features already encode digits, phrases, brands and field overlap; they do not explicitly enforce a general rule that a requested model, size or exclusion must agree with the product. A close lexical match can therefore coexist with an incompatible item or accessory. This motivates representing explicit requirements and exposing the learner to difficult, human-judged negatives that differ in a decisive attribute. The completed experiments also constrain this diagnosis: the five added constraint features yield a small XGBoost gain below the promotion threshold, while the tested hard-negative weighting does not improve the selected result. A future study should distinguish

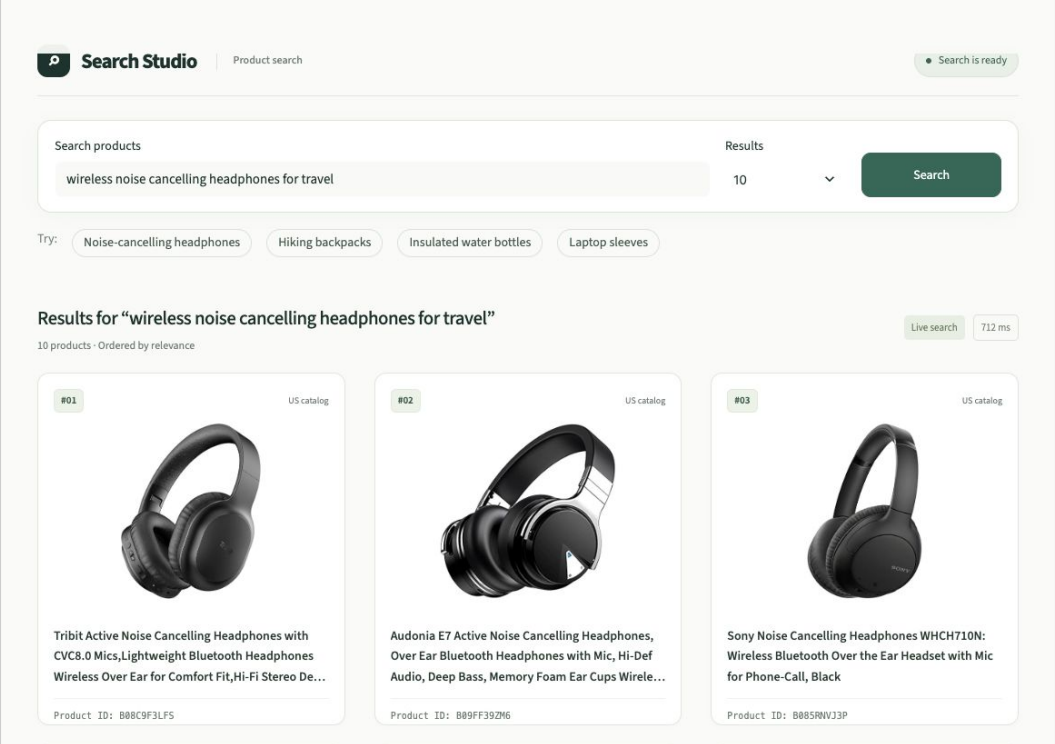


Figure 6: The actual English interface calling FastAPI and displaying real catalog products. Visible query time illustrates one interaction; performance claims use separate measured workloads.

verified requirement conflicts from uncertain or absent metadata, and test better-covered constraint extraction and curated near-match negatives rather than assume that stronger penalties alone will solve the errors.

6.3 Week3: Intent Preservation Before Query Rewriting

Short and nonstandard queries can leave intent genuinely underspecified. The saved examples include “newborn baby boy bunjee” and “truly’s”, for which spelling, entity identity and product type cannot all be established from the query alone. Query rewriting should preserve supported intent rather than invent a model number, size or desired product. Supervised fine-tuning (SFT) on reviewed query–intent–rewrite examples is a possible next step, with held-out entities and product types needed to assess generalization. Feedback-based reinforcement learning could then be investigated if a reliable relevance reward and safeguards against intent drift are available. Prior work demonstrates feedback-trained rewriting in question-answering retrieval, providing a methodological precedent rather than evidence of a gain in this product-search system [16]. Online reinforcement learning would additionally require real interaction feedback and a separately evaluated policy. Neither SFT rewriting nor online reinforcement learning is completed here, and neither is enabled in the frozen pipeline.

6.4 Week5: Fine-Grained Query–Product Interaction

The ranker receives 28 scalar features, not the complete query and product text. These features summarize substantial lexical and semantic evidence, but may not expose the exact relationship that decides relevance. For “toddler sports bedding”, an I-labeled twin bedspread displaces an E-labeled toddler set; for “stand mixer”, the saved Cold comparison instead shows a useful shift from a cover toward the appliance. Negated queries such as “without light” and “not dehydrator” expose another difficulty, although conflicting title cues and released labels prevent a purely title-based correctness judgment. High embedding similarity alone therefore provides no guarantee that each fine-grained requirement is met. A cross-encoder offers a way to score jointly encoded query and product tokens rather than only independent embedding similarity [4]; this also explains the rationale

for the historical cross-encoder approach. Its value on the current benchmark would need a new controlled study. The unsuccessful field-feature and loss experiments show that the tested lightweight changes do not yet close this gap.

6.5 Week6: Unseen Products, Missing Images, and Calibration

The weak Cold improvement is consistent with limited generalization to unseen product IDs, but the experiments do not isolate a single cause. Queries and catalog descriptions express needs differently, whereas the projection objective aligns product text with product images; improved training loss need not improve query relevance. The adapted model also loses held-out alignment quality and catalog recall, supporting retention of the frozen representation. Missing images introduce a separate, directly observed mechanism: raw text fallback and text–image fusion have different score distributions. Train-only calibration and an availability mask address that original bias, although the recorded missing-image recall tradeoff remains. Missing-image products are not uniformly worse in the final ranking: their small Cold subgroup has a higher NDCG point estimate than the image-available subgroup, over different eligible queries. More diverse queries, specification-level negatives, product types and image-availability conditions are reasonable future training directions, to be tested with protected product holdouts. Diversity is a proposed intervention, not an established explanation or guaranteed remedy for the remaining Cold errors.

6.6 Evaluation Decisions

Hard-case inspection identifies useful questions without replacing the paired aggregate results. The archived cases are selected extremes, so they do not estimate how common each failure type is. The final test supports modest Warm gains, while the ListNet–LambdaMART interval does not establish stable algorithm superiority. RRF is promising across recall comparisons, but its approved Cold-Val promotion test remains uncertain, so it stays auxiliary. Known-positive recall and fixed-pool NDCG answer different questions; better candidate coverage does not satisfy the Week5 or Week6 ranking targets. These decisions remain fixed while the proposed annotation, constraint and rewriting improvements belong to future experiments.

6.7 Efficiency and User Experience

Large speedups arise from preserving computations and changing when data are prepared. A small MLP is not necessarily the service bottleneck when each request repeatedly indexes and converts a large product table. Prepared static state and exact row-key mapping remove that overhead without changing the selected model. The evidence nevertheless has three boundaries: historical same-session bitwise equality, cross-environment floating-point compatibility, and end-user HTTP behavior. Near-tie order changes can occur across environments even when the Top10 remains stable. Likewise, low warmed sequential latency and successful 64-client queuing are different properties. The final demo supports responsive individual use. Broader repeated loads expose timeouts and query-supply exhaustion, so bounded admission must not be presented as stable error-free 64-client capacity or production readiness.

7 Limitations and Future Work

Relevance coverage and benchmark scope. The catalog has only partial human judgments. KP-R estimates coverage of observed positives, and unknown results can be useful despite having no label. The custom source subset, gain mapping, grouped split, and candidate masks also prevent comparison with official ESCI challenge scores. Historical synthetic queries and silver labels have additional judge dependence; their limited human review does not convert them into a human-gold ranking benchmark. A useful future evaluation would collect independently judged, sufficiently complete pools and assess visual relevance directly.

Cold definition and uncertainty. Product-ID holdout is not real zero-interaction cold start. Duplicate-content diagnostics reduce specified overlap but do not audit all semantic duplicates or pretraining exposure. Small missing-image Cold samples and differing eligible-query sets limit subgroup inference. Query bootstrap intervals omit training-seed variability and repeated-development selection uncertainty, and multiple reported comparisons are not multiplicity-adjusted. A new temporal or

interaction-grounded study would require new protected data; the completed Final Test must not be reused for further tuning.

Model and source identity. Exact historical SQID CLIP revision and preprocessing identity remain unresolved. The live service uses an explicitly approved, pinned compatibility-tested encoder, with small vector errors and some lower-rank changes. Cross-CPU bitwise reproduction is also not universal. The observed approximately 10^{-7} document-length feature differences are consistent with floating-point execution differences, but the exact instruction-level cause has not been established. The complete primary artifact for the older 48K headline and some historical query-understanding acceptance records are unavailable. These gaps are preserved rather than filled by reconstruction claims.

Serving and deployment. Local repeated evaluation separates Hot64 and Broad1000, Burst and sustained closed-loop traffic, and empty versus two-pass conditioned product caches. Four A/B/C repeats and only two complete Broad sustained windows limit inference about long-running stability. Early terminations, HTTP timeouts and the interrupted fifth-repeat plan remain reported. Mean core latency below 200 ms does not establish the same P95 bound or saturated HTTP latency. No remote browser/network, image-delivery or production-availability measurement is completed. Dynamic cross-request batching, a validated GPU online path and public hosting remain future work. A Dockerfile exists, but its final dual-environment build and launch have not been accepted.

Remaining integration scope. The final ESCI service does not implement and validate insertion of newly supplied products into the complete catalog index. An earlier historical subset-index insertion demo is separate evidence. The original LLM query-understanding objective likewise remains outside the final formal path; historical implementations and partial evaluations do not establish the missing complete acceptance result. Further work should begin from clearly specified data and identity requirements, rather than assuming that adding these components will improve relevance.

8 Conclusion

The final system combines frozen SBERT/CLIP representations, calibrated exact retrieval, and ListNet ranking. On the human-judged benchmark, fusion improves Warm known-positive Recall@50 by 14.35%; ranking gains are smaller and do not meet the Week2, Week5, or Week6 NDCG targets. Projection adaptation and the controlled ranking extensions are not adopted, while RRF remains a registered auxiliary flow.

Equivalent preparation and candidate mapping preserve predictions while substantially reducing inference overhead. Repeated local CPU measurements average 102.16 ms for the cached core, 164.03 ms with live encoding and 184.52 ms over serial HTTP. The natural-language demo is functional, but tail-latency variation and 64-client failures limit stronger serving claims. These findings support calibration and careful inference engineering, while Cold generalization and production capacity remain unresolved.

Hard-case inspection motivates more reliable annotation, explicit product requirements, and intent-preserving query representations as future directions. The completed constraint, loss and projection experiments also show that plausible diagnoses do not automatically yield an effective improvement.

References

[1] Chandan K. Reddy, Lluís Màrquez, Fran Valero, Nikhil Rao, Hugo Zaragoza, Sambaran Bandyopadhyay, Arnab Biswas, Anlu Xing, and Karthik Subbian. Shopping queries dataset: A large-scale ESCI benchmark for improving product search, 2022. URL https://arxiv.org/abs/2206.06588. arXiv preprint arXiv:2206.06588.

[2] Marie Al Ghossein, Ching-Wei Chen, and Jason Tang. Shopping queries image dataset (SQID): An image-enriched ESCI dataset for exploring multimodal learning in product search. In *Proceedings of the ACM SIGIR Workshop on eCommerce 2024*, volume 3843 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2024. URL https://ceur-ws.org/Vol-3843/paper_17.pdf.

[3] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019.

[4] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1410.

[5] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

[6] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.

[7] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016. doi: 10.1145/2939672.2939785.

[8] Qiang Wu, Christopher J. C. Burges, Krysta M. Svore, and Jianfeng Gao. Adapting boosting for information retrieval measures. *Information Retrieval*, 13(3):254–270, 2010. doi: 10.1007/s10791-009-9112-1.

[9] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html.

[10] Zhe Cao, Tao Qin, Tie-Yan Liu, Ming-Feng Tsai, and Hang Li. Learning to rank: From pairwise approach to listwise approach. In *Proceedings of the 24th International Conference on Machine Learning*, pages 129–136. ACM, 2007. doi: 10.1145/1273496.1273513.

[11] G. V. Cormack, C. L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. ACM, 2009. URL https://plg.uwaterloo.ca/~gvcormac/cormacksigir09-rrf.pdf.

[12] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021. doi: 10.1109/TBDATA.2019.2921572.

[13] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

[14] Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002. doi: 10.1145/582415.582418.

[15] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023. doi: 10.52202/075280-2020.

[16] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. Query rewriting in retrieval-augmented large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5303–5315. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.322.

A Data Lineage, Splits, and Metric Implementation

A.1 Keys, Content, and Split Construction

The canonical product key is the pair (locale, product ID); query IDs remain strings. Frozen product and embedding-row maps join the ESCI US subset to SQID. Product text for SBERT and lexical features concatenates title, brand, bullet point, description, and color with newlines. CLIP product-text features use the SQID title representation. Prices, ratings, counts of reviews, and product categories are unavailable. Product occurrence bins are used for split stratification only; occurrence counts, frequency bins, role labels, IDs, and relevance-derived variables are not ranking features.

The query grouping rule first applies NFKC normalization, case folding, and whitespace normalization. Exact normalized queries and equal token sets are grouped. A fixed near-duplicate test uses character 3–5-gram cosine similarity at least 0.92, token Jaccard at least 0.8, and equal digit sets. Connected grouping precedes deterministic 70/15/15 allocation. Products are stratified by occurrence bins 1, 2, 3–4, and 5+, crossed with image availability, and allocated 80/10/10 by seeded hashing/shuffling. The data audit finds no cross-role ID overlap or near-duplicate query edges under the registered rule. This does not prove the absence of all semantically related queries or duplicate product content.

Training views are explicit. Pointwise learning retains all Train/Warm labels, even when a query loses its positives after the Warm filter. The LTR view requires at least one E/S product, at least two candidates, and at least two distinct grades. It removes 98 queries and 1,128 pairs relative to the pointwise view. All three Week5 model families use this same LTR view. Lexical corpus statistics and pretrained representations may use unlabeled full-catalog content, including Cold content; Cold labels and Cold alignment pairs do not enter supervised or projection training. This is content-available generalization, not a strictly inductive prohibition on all held-out product content.

The content-unique Cold diagnostic excludes target products with registered normalized-title, image-URL, or exact image-vector overlap across protected roles. It does not rebuild the catalog or remove competing products from the ranked list. Alignment training has its own content filter: 89,751 eligible Warm-Train pairs become 88,063 after excluding 1,688 overlaps; 17,394 Warm-Val alignment pairs become 16,853 after excluding 541. These filters are not a claim that all benchmark products are content-disjoint.

A.2 Scoped Ranking and Eligibility

A non-target candidate remains at its predicted position and contributes zero target gain. For example, if a Warm competitor precedes the first Cold E item, that item retains its lower rank in the Cold metric. Deleting the Warm competitor would answer a different, easier within-Cold question. The report excludes an older diagnostic that dropped such competitors even though it was labeled as a Warm-only result.

Let $b_r = \mathbb{1}[p_r \in \mathcal{P}_q^S]$, $P_r = \sum_{j \le r} b_j / r$, and $n_q = |\mathcal{P}_q^S|$. Then

$$\mathrm{AP}_{q,S}@10 = \frac{\sum_{r=1}^{\min(10,|\mathcal{J}_q|)} b_r P_r}{\min(10, n_q)}, \qquad \mathrm{RR}_{q,S} = \begin{cases} 1/\min\{r : b_r = 1\}, & \exists r : b_r = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Only queries with $n_q > 0$ enter a scope's aggregate. An E/S eligibility threshold does not discard C gains from NDCG: target C retains gain 1. Catalog coverage is $K^{-1} \sum_{r \le K} \mathbb{1}[p_r \in \mathcal{J}_q \cap S]$ for a full Top$K$ return. Coverage and known-positive recall have different denominators and should not be compared as the same quantity.

A single query can be eligible for both image and missing-image targets, or both Train-Seen and Train-Unseen targets. Query counts in Table 16 therefore need not add across rows. Overall targets Warm ∪ Cold-Test, a catalog target set of 148,410 products, while retrieval still searches all 164,900 products. All 1,343 Test queries receive predictions; scope eligibility affects only subsequent metric averages.

A.3 Prediction Lock and Verification Scope

Before label unlock, each fixed-pool model has 27,487 predicted candidate positions and each catalog flow has 134,300 output positions. The lock covers prediction contents and candidate keys together with code and configuration identity. The saved independent review checks 1,270 prediction-lock

Table 16: Final Test scope-specific eligibility and target populations.

Target scope	Eligible n	E/S pairs	Judged pairs	Catalog products
Warm	1,337	17,437	22,042	131,920
Cold-Test	1,012	2,208	2,770	16,490
Overall	1,339	19,645	24,812	148,410
Warm / image	1,332	16,649	21,026	124,880
Warm / no image	369	788	1,016	7,040
Cold-Test / image	979	2,114	2,644	15,610
Cold-Test / no image	86	94	126	880
Overall / image	1,333	18,763	23,670	140,490
Overall / no image	397	882	1,142	7,920
Cold-Test / content unique	982	2,085	2,627	15,686
Content unique / image	949	1,995	2,510	14,897
Content unique / no image	82	90	117	789
Week2 Warm / Train-Seen	657	2,016	2,766	94,768
Week2 Warm / Train-Unseen	1,333	15,421	19,276	37,152
Week5 Warm / Train-Seen	653	2,000	2,749	93,788
Week5 Warm / Train-Unseen	1,333	15,437	19,293	38,132

Notes. All scopes start from the same 1,343 Final Test queries and retain all competitor positions. Cold-Val products remain masked competitors. Image groups describe target-product embedding availability. Content-unique Cold-Test excludes protected content overlaps. Train-Seen membership depends on the original Week2 pointwise versus Week5 LTR training view; these are distinct populations. Counts do not imply exclusive query sets.

files, 193 completion artifacts, and 43 registered dependencies. It reproduces 1,573,203 saved metric scalars with maximum absolute difference 4.44×10^{-16} and checks 162 paired comparisons. Large Drive assets were hash-verified during the recorded Colab execution; report preparation does not independently reread every remote large file. Native-model tolerance replay has zero metric impact in that final review. These checks concern the released execution and are not a second final model-selection run.

B Original Feature Contract and Calibration

B.1 Lexical Definitions and Precision

Custom field tokenization uses the Unicode pattern `(?u)\b\w+\b` with case folding. Let Q be the unique query-token set, T_f the unique product tokens for field f, and D the concatenated document. Define $\mathrm{cov}(Q,T) = |Q \cap T| / \max(1,|Q|)$. Phrase checks retain token order and occurrences, whereas coverage uses sets. The CountVectorizer uses its frozen lowercasing/tokenization and a maximum of 150,000 vocabulary entries; it is not replaced by the custom token-set code.

With N catalog documents, retained-vocabulary count $\mathrm{tf}_{u,p}$, document length ℓ_p, and corpus mean $\bar{\ell}$, the frozen BM25 feature is

$$\mathrm{BM25}(q,p) = \sum_{u \in Q_V \cap V} \log\left(1 + \frac{N - \mathrm{df}_u + 0.5}{\mathrm{df}_u + 0.5}\right) \frac{\mathrm{tf}_{u,p}(k_1+1)}{\mathrm{tf}_{u,p} + k_1(1 - b + b\ell_p/\bar{\ell})}. \tag{11}$$

The constants are $k_1 = 1.5$ and $b = 0.75$. Here V is the frozen CountVectorizer vocabulary and the BM25 query sum uses its lowercased query-token set, denoted Q_V, rather than the custom case-folded coverage set Q. Query terms contribute once. Document length sums retained-vocabulary counts, not all raw tokens. TF–IDF uses raw term counts, smoothed IDF $\log((1+N)/(1+\mathrm{df})) + 1$, and L2 normalization, with no sublinear TF. Title TF–IDF uses the same vocabulary and document-fitted IDF, not a separately fitted title corpus.

Every output feature column is float32 in the stated order. Sparse corpus matrices, embedding arithmetic, calibration intermediates, and log-length operations retain their original conversion points; in particular, calibration uses float64 intermediates before the final feature cast. Prepared inference caches values at those same points. Scaler fitting and application use the frozen training moments and original float64-to-float32 path. A numerically close replacement formula is not assumed equivalent merely because it has the same conceptual meaning.

Table 17: Complete original 28-feature order and definitions. Q = query only, P = product only, QP = query–product. Field-empty means no tokens after the original missing-value handling.

No.	Frozen name	Definition / branch	Input
1	`bm25`	Frozen BM25 score defined above.	QP
2	`tfidf`	L2-normalized query/document TF–IDF inner product.	QP
3	`title_tfidf`	Query/title TF–IDF inner product with shared vocabulary and IDF.	QP
4	`query_coverage`	$\mathrm{cov}(Q, T_D)$.	QP
5	`title_coverage`	$\mathrm{cov}(Q, T_{\mathrm{title}})$.	QP
6	`brand_overlap`	$\mathrm{cov}(Q, T_{\mathrm{brand}})$.	QP
7	`color_overlap`	$\mathrm{cov}(Q, T_{\mathrm{color}})$.	QP
8	`bullet_coverage`	$\mathrm{cov}(Q, T_{\mathrm{bullet}})$.	QP
9	`description_coverage`	$\mathrm{cov}(Q, T_{\mathrm{description}})$.	QP
10	`digit_coverage`	Coverage of $Q_d = \{u \in Q : u \text{ contains a digit}\}$ in T_D, denominator $\max(1, \lvert Q_d \rvert)$.	QP
11	`has_digits`	$\mathbb{1}[\lvert Q_d \rvert > 0]$.	Q
12	`title_phrase_match`	Nonempty ordered tokenized query is a space-bounded contiguous phrase in tokenized title.	QP
13	`brand_phrase_match`	Nonempty complete tokenized brand is a space-bounded phrase in tokenized query.	QP
14	`query_length`	$\log(1 + \lvert Q \rvert)$, unique tokens.	Q
15	`title_length`	$\log(1 + \lvert T_{\mathrm{title}} \rvert)$, unique tokens.	P
16	`document_length`	$\log(1 + \ell_p)$ using the frozen retained-vocabulary count length.	P
17	`missing_description`	$\mathbb{1}[\lvert T_{\mathrm{description}} \rvert = 0]$.	P
18	`missing_bullet`	$\mathbb{1}[\lvert T_{\mathrm{bullet}} \rvert = 0]$.	P
19	`missing_brand`	$\mathbb{1}[\lvert T_{\mathrm{brand}} \rvert = 0]$.	P
20	`missing_color`	$\mathbb{1}[\lvert T_{\mathrm{color}} \rvert = 0]$.	P

Continued on next page

Table 17 (continued)

No.	Frozen name	Definition / branch	Input
21	`sbert_similarity`	$s_q^\top s_p$.	QP
22	`clip_text_similarity`	$c_q^\top t_p$.	QP
23	`clip_image_similarity`	$c_q^\top i_p$ if an image vector is valid, otherwise zero.	QP
24	`image_available`	m_p; this is embedding availability, not a downloaded-image check.	P
25	`calibrated_text_fusion_score`	$0.5s + 0.5t$. Despite its frozen name, this uses raw text cosine, without affine calibration.	QP
26	`calibrated_fusion_score`	$0.5s + 0.5C_{0.5}(q, p)$ from Eq. (4), mask-aware.	QP
27	`brand_length`	$\log(1 +$ number of brand-token occurrences$)$, not set size.	P
28	`color_length`	$\log(1 +$ number of color-token occurrences$)$, not set size.	P

The ablation dependency declaration assigns brand/color overlap and brand phrase matching to Metadata. Semantic contains SBERT, CLIP text, and raw text fusion; Image contains masked image similarity and availability. Calibrated fusion depends on both Semantic and Image and is removed by either ablation. Query length remains Lexical; title/document/brand/color lengths and all missing flags are Metadata; other query/field matching columns are Lexical. This dependency policy explains why group effects overlap.

B.2 Calibration Values and Affine Coefficients

The fixed unlabeled sample has 512 Train queries, 128 Warm products per query, 62,046 image-available score pairs, and 3,490 missing-image pairs. Its stored parameters are

$$\begin{aligned} \mu_t &= 0.22564778817401995, & \sigma_t &= 0.1249746794951641, \\ \mu_i &= 0.09940735359384505, & \sigma_i &= 0.041104024512949464, \\ \mu_m &= 0.2945886025676253, & \sigma_m &= 0.1612383415483866, \\ \rho &= 0.037969082622833036. \end{aligned}$$

For image-available products, $a_p = 1 - \beta$, $b_p = \sigma_t\beta/\sigma_i$, and $d_p = \beta\mu_t - \sigma_t\beta\mu_i/\sigma_i$. For missing-image products, $a_p = \sigma_t v_\beta/\sigma_m$, $b_p = 0$, and $d_p = \mu_t - a_p\mu_m$. These coefficients yield Eq. (5); they are not learned relevance weights. The extra constant coordinate carries the branch offset. All 164,900 products remain searchable.

Exact search requests an additional neighbor to detect a cutoff tie. A range search expands a tied boundary using the original nextafter threshold, then score and canonical row/product-key order determine the final Top100. Feature/ranking ties retain the original locale/product-ID ordering. Candidate mapping optimization substitutes aligned arrays for expensive row materialization, not a different tie rule.

C Frozen Configurations and Reproduction Assets

The original Week2 XGBoost uses Warm Validation RMSE stopping and saves 77 trees (best iteration 76, zero based). This is distinct from Week5's Warm NDCG-based checkpoint checks and from P1's explicit NDCG stopping alternative. In the original Week5 run, selected checkpoints are XGBoost 60 trees, LambdaMART 180 trees, and ListNet epoch 6. Tree budgets are 600 rounds, checked every 20, with five non-improving checks; ListNet has a 30-epoch maximum and patience five. The original model seed is 20261014 and numeric thread count is four.

The SBERT source is `sentence-transformers/all-MiniLM-L6-v2`, revision `1110a243fdf4706b3f48f1d95db1a4f5529b4d41`. Its live path uses attention-mask mean pooling, maximum length 256, and L2 normalization. The compatibility-approved CLIP source is `openai/clip-vit-large-patch14`, revision `32bd64288804d66eefd0ccbe215aa642df71cc41`, with text projection, maximum length 77, and L2 normalization. Its registered weight SHA256 begins `a2bf730a0c7debf1`; the full hash and tokenizer file hashes are preserved in the source evidence map. This identifies the tested live artifact, not the unrecorded historical SQID extractor revision.

Dependency	Registered SHA256 prefix	Verification record
Query/product split manifest	`0597900737dc9161`	Local and final execution
Original Week2 lexical index	`32bfee55efb5c237`	Local and final execution
Formal mask-aware FAISS index	`70062f0e1e005228`	Final Colab execution
Formal Week5 lexical index	`442e9336b3d4a816`	Final Colab execution
Original LambdaMART weights	`377f1ed6e15ea5f5`	Local and final execution
Original ListNet NumPy weights	`b4e70b6d09dacc03`	Local and final execution

Full paths, identities, and registration-stage versus execution verification records are preserved in the machine-readable evidence. Prefixes identify files compactly and do not replace full SHA256 verification.

The Week2 and Week5 lexical indices must not be interchanged. P0 finds 3,756 distinct frequency-one vocabulary terms on each side and 5,929 different retained document lengths despite matching high-level settings. Their exact originating machine/runtime cause is unresolved. Subsequent work fixes the Week5 index, vocabulary, feature order, and dtype rather than overwriting either historical artifact. The frozen dependency map records full SHA256 values for the product/query keys, split

Table 18: Frozen model configuration and selected development checkpoints.

Component	Saved configuration / selection
Lexical index	BM25 $k_1 = 1.5$, $b = 0.75$; 150,000 maximum TF–IDF features
Original Week2 XGBoost	Squared-error grade regression; depth 6, LR 0.05, L2 10; best iteration 76 (zero based)
Frozen SBERT	sentence-transformers/all-MiniLM-L6-v2; 384 dimensions; max length 256; float32, L2 normalization
Calibration	Train-only 512 queries $\times$ 128 products; selected $\alpha = \beta = 0.5$
Controlled XGBoost (28)	Squared-error; depth 6, LR 0.05, L2 10; selected 60 trees
Controlled LambdaMART	Lambdarank gains [0, 1, 3, 7]; 31 leaves; LR 0.05; truncation 10; selected 180 trees
Final ListNet	MLP $28 \to 64 \to 32 \to 1$; grade-softmax top-one loss; Adam LR 0.001; decay 1e-05; 64 queries/batch; epoch 6
Selection budget	Trees: 600, checked every 20; ListNet: 30 epochs; patience 5; Warm Validation NDCG@10
Formal retrieval	Exact FlatIP, 164,900 products $\times$ 1,153 dimensions; Top100; mask-aware affine calibration
Frozen Scaler / features	Original Train-fit mean/std in float64; standardized outputs in float32; 28 original features; neither Scaler nor ranker re-estimated for RRF60

Notes. Configurations and checkpoints were frozen before Final Test. Week2 and Week5 lexical/model artifacts are distinct. Model selection precedes final Test. Seed 20261014 applies to these original training stages; optimization protocols use their separately saved folds/seeds.

manifest, feature contract, scaler, models, embeddings, and index. The split-manifest SHA begins `0597900737dc9161`.

Original model builders are in `project/esci_sqid/src/week2_workflow.py`, `week5_models.py`, and `week6_alignment.py`; retrieval and calibration are in `week4_vectors.py` and `week4_calibration.py` under the same source directory. The final executor is `project/esci_sqid/final_test_v1/`. Engineering implementations are isolated in `project/esci_sqid/week7/`, `week7_p2/`, `week7_online/`, and `week7_queue64/`. Historical metadata code remains under `project/final100k_src/`. These are reproduction entrypoints into preserved assets, not instructions to rerun a sealed final evaluation for model selection.

The accompanying report source includes exact table pointers and hashes, bibliography verification, figure inputs, the approved evidence matrix, and a build script. It deliberately excludes model weights, catalog data, raw relevance labels, and access credentials. The audit classifies 180 claims: 126 verified/completed, four implemented but incompletely evaluated, 33 experimental/not adopted, six proposed/not implemented, and 11 blocked/unresolved. This report places completed methods and outcomes in the main narrative, evaluated alternatives in Results, and unresolved or unimplemented items in Discussion/Future Work.

D Controlled Optimization Details and Curves

D.1 Matched Comparisons and Budgets

P1 uses only Train query groups for a fixed 80/20 internal split, preserving the Week2 and LTR training views separately. XGBoost keeps squared-error regression while testing query-level NDCG stopping, query-balanced sample weights, and a factor-two weight for existing C/I labels whose BM25 is at least their own query's 75th percentile and whose query coverage is at least 0.75. The latter starts from equal-query weights, renormalizes each query total, then rescales the global mean weight to one. LambdaMART compares its original settings with L2=5 and a 15-leaf/minimum-leaf-count-40 alternative. Other inputs remain fixed within each contrast. Internal point differences are not promoted when their uncertainty or stability is insufficient.

P1 and P2 use split seed 20261015. P2 fixes three query-group folds, with 20% of available query groups held out for inner checkpoint selection, followed by fixed-budget refitting inside each outer training partition. Every new model has its own matched retrained baseline on those same folds. The XGBoost constraint comparison uses the original 20 features versus 25; LambdaMART and ListNet compare 28 versus 33. The expanded ListNet preserves the 64/32 hidden architecture and initializes additional input columns to zero. This differs from comparing all three algorithms on an identical feature count. P2 feature rules are generic text parsers, with uncertainty treated neutrally, rather than exceptions written for external Validation queries.

The five constraint columns are model-number exact match, numeric-constraint conflict, product-type match, accessory relation, and negation conflict. The later field-level study separately examines title/description coverage, brand agreement/conflict, numeric units, and main-product/accessory evidence. Saved feature-support analyses, parser tests, and examples are retained with the experiment records. A small support-dependent feature effect is not assumed to apply to every query. Neither feature family enters the final original28 contract.

D.2 Losses and Product Holdout

Gain-aware ListNet uses $P_y(j) = (G(y_j) + 0.1)/\sum_k (G(y_k) + 0.1)$, the same 28 inputs, optimizer, and maximum epoch budget as its baseline. This gives a uniform target for an all-zero-gain valid list; padding is excluded from normalization. The later position-aware loss sorts current detached scores and defines $D(r) = 1/\log_2(r+1)$ for $r \le 10$, zero otherwise. For permitted pairs with $y_i > y_j$,

$$w_{ij} = \frac{|(G(y_i) - G(y_j))(D(r_i) - D(r_j))|}{\text{IDCG@10}}, \qquad \mathcal{L}_q = \sum_{i,j:y_i>y_j} w_{ij} \log(1 + e^{-(f_i - f_j)}). \tag{12}$$

The implementation averages over active query losses, without normalizing each query by the sum of pair weights; its fixed logistic scale is one. Zero-IDCG lists or lists with no active pair contribute no invalid division or gradient. This is the specific tested surrogate, not an assertion of equivalence to every published LambdaLoss formulation.

The pseudo-Cold split uses product-hash seed 20261016 and holds out 18,758 of 93,788 supervised Warm product IDs (20%), excludes 20,202 supervised pairs including 15,912 E/S pairs from both matched models, and leaves content available for inference. Query-group separation remains in force. Warm and Overall safety metrics are computed with the complete candidate competition order, and the average fraction of remaining Warm evaluation pairs whose products are actually Train-Seen is only about 9.12%. These populations differ from both the fixed external Cold-Val partition and genuine zero-click products.

For P2 and the targeted ranking directions, promotion requires an average absolute NDCG improvement of at least 0.002, a positive paired interval lower bound, and no single-fold NDCG delta below -0.002. The targeted pseudo-Cold safety gate also requires Warm/Overall paired-interval lower bounds at least -0.002. A positive interval alone does not meet the practical-effect gate. Fold-level differences reveal instability hidden by pooled point estimates: the position-aware Warm objective is positive in only one fold, while its pseudo-Cold effect is negative in all three. The RRF pseudo-Cold recall contrast is positive in all three, but measures a different endpoint. Query-paired bootstrap intervals treat models/folds as fixed and do not capture the full dependence induced by retraining within folds.

The saved Week5 curves show selected LambdaMART checkpoint 180 and ListNet epoch 6 followed by five lower Validation checks, matching the stopping rule. ListNet loss ends lower while Validation remains below its selected maximum; this supports early stopping but does not distinguish overfitting from Validation noise. LambdaMART training loss was not saved and is not drawn. Across the five projection epochs, loss decreases and alignment improves within the adapted family, but every checkpoint remains below the frozen alignment reference.

D.3 Recovery and Verification Boundaries

The targeted run's original tail remained silent after retrieval batch 97/97 and was recovered from saved intermediate artifacts. Reports preserve the recovery provenance and reuse identity-checked completed work; they do not count recovery as a new independent replication. P1/P2/targeted configurations and unsuccessful outcomes remain archived. The report's figure and table builders read only these saved aggregates/histories. They do not load raw test relevance, perform inference, or change the formal model.

E Final Subgroup Diagnostics

Tables 20 and 21 expand the final scopes. Image availability is defined by a usable SQID embedding, not by successful image rendering in a later browser. Content-unique and Train-Seen targets change only relevance masks and eligibility; the same predicted competitor order is retained. Their point values should not be read as interchangeable estimates for a single fixed query population.

Table 19: Fold-level deltas and the distinction between fold means and query macros.

Candidate contrast	Fold 1	Fold 2	Fold 3	Fold mean	Query macro	Positive
Constraints / XGB	+0.00164	+0.00317	+0.00046	+0.00176	+0.00176	3/3
Constraints / LambdaMART	+0.00310	-0.00072	+0.00225	+0.00154	+0.00154	2/3
Constraints / ListNet	+0.00179	+0.00116	+0.00114	+0.00137	+0.00136	3/3
Gain-softmax / ListNet	+0.00011	-0.00069	-0.00128	-0.00062	-0.00062	1/3
Position-aware loss	-0.00083	-0.00067	+0.00300	+0.00050	+0.00050	1/3
Field features / LambdaMART	+0.00146	-0.00022	-0.00008	+0.00039	+0.00039	1/3
Field features / ListNet	-0.00080	-0.00065	-0.00027	-0.00057	-0.00057	0/3
Position loss / pseudo-Cold	-0.00417	-0.00370	-0.00112	-0.00300	-0.00301	0/3
Clean RRF pseudo-Cold KP-R@50	+0.02013	+0.02232	+0.02484	+0.02243	+0.02241	3/3

Notes. All rows except the last use NDCG@10; the final row uses Known-Positive Recall@50. The fold mean gives each fold equal weight, whereas the query macro gives each eligible query equal weight; use the query-macro delta with the paired query-bootstrap interval in Tables 9 and 6. The folds are shared query groups with inner checkpoint selection/refit.

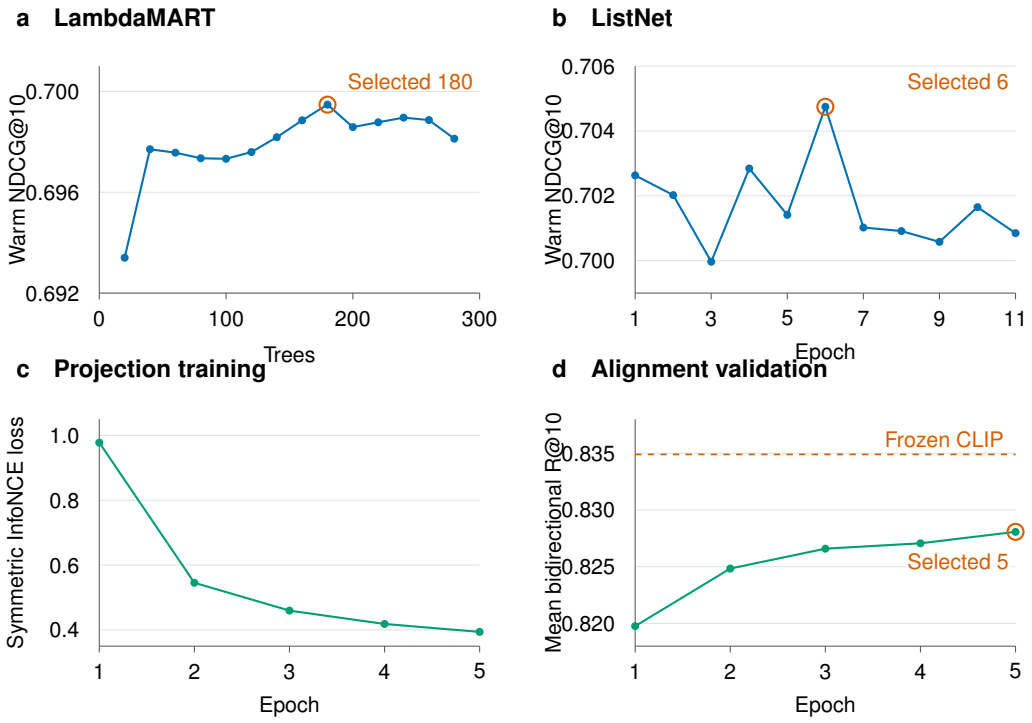


Figure 7: Saved development training trajectories. Axes and protocols are kept separate; checkpoints are chosen within their permitted development roles. Selected checkpoints are circled and frozen CLIP alignment is dashed; each panel has its own units and scale. Reduced training loss is not by itself evidence of improved held-out alignment or ranking. No curves are generated by new model execution.

Table 20: Final fixed-pool subgroup diagnostics.

Target scope	n	BM25	Week2 XGB	LambdaMART	ListNet	LN MAP	LN MRR
Warm / image	1,332	0.6452	0.6520	0.6872	**0.6875**	0.5857	0.8110
Warm / no image	369	0.3591	0.3697	0.3883	**0.4252**	0.3118	0.3917
Cold-Test / image	979	0.3059	0.3179	**0.3248**	0.3145	0.1984	0.2924
Cold-Test / no image	86	0.2840	0.2792	0.3477	**0.3902**	0.2865	0.3168
Overall / image	1,333	0.6992	0.7088	**0.7459**	0.7442	0.6851	0.8647
Overall / no image	397	0.3606	0.3713	0.3973	**0.4375**	0.3230	0.4086
Cold-Test / content unique	982	0.3107	0.3182	**0.3288**	0.3209	0.2051	0.2994
Content unique / image	949	0.3080	0.3169	**0.3221**	0.3110	0.1966	0.2888
Content unique / no image	82	0.2743	0.2790	0.3371	**0.3864**	0.2824	0.3132
Week2 Warm / Train-Seen	657	0.2749	0.2823	**0.2869**	0.2850	0.1746	0.2932
Week2 Warm / Train-Unseen	1,333	0.6315	0.6387	0.6777	**0.6817**	0.5698	0.8094
Week5 Warm / Train-Seen	653	0.2729	0.2805	**0.2854**	0.2828	0.1723	0.2910
Week5 Warm / Train-Unseen	1,333	0.6318	0.6389	0.6781	**0.6824**	0.5709	0.8103

Notes. The four model columns report NDCG@10; the last two report ListNet MAP@10 and untruncated MRR. Bold marks the highest NDCG point estimate within a row only. Each row retains all competitors and masks non-target relevance; no Warm-candidate-only diagnostic is used. Small no-image Cold samples and changing eligibility limit comparisons. Saved uncertainty remains in the machine-readable source map/report; the primary-scope intervals are in Table 2.

Table 21: Formal and auxiliary catalog flow subgroup diagnostics.

Target scope	n	A R@50	B R@50	A R@100	B R@100	A Cov.	B Cov.
Warm / image	1,332	0.5833	0.6004	0.6715	0.6827	0.1636	0.1687
Warm / no image	369	0.2521	0.5836	0.2557	0.6285	0.0118	0.0264
Cold-Test / image	979	0.5736	0.5862	0.6604	0.6757	0.0272	0.0276
Cold-Test / no image	86	0.3430	0.6105	0.3430	0.6686	0.0077	0.0142
Overall / image	1,333	0.5817	0.5982	0.6694	0.6812	0.1842	0.1896
Overall / no image	397	0.2588	0.5859	0.2622	0.6344	0.0127	0.0278
Cold-Test / content unique	982	0.5616	0.5879	0.6460	0.6795	0.0265	0.0275
Content unique / image	949	0.5734	0.5887	0.6624	0.6814	0.0268	0.0272
Content unique / no image	82	0.3232	0.5915	0.3232	0.6524	0.0071	0.0137

Notes. R denotes Known-Positive Recall; Cov. denotes target judged coverage@50. A is Fusion–ListNet; B is RRF60–same ListNet. Both use Top100 and all product competitors. Target subsets and eligibility match Table 16; unjudged products remain unknown. These are subgroup diagnostics, not new selection criteria.

The image-available Cold subset has 2,114 known E/S pairs over 979 eligible queries, while the missing-image subset has 94 over 86 queries. A query may appear in both. The content-unique Cold subset has 15,686 catalog products and 2,085 known E/S pairs over 982 eligible queries. Week2 and Week5 Train-Seen definitions differ because their supervised views differ. In neither case is supervision equivalent to measured real user clicks. Full machine-readable intervals and companion metrics are included in the archived metric export and the report source evidence ledger; no additional subset is selected after seeing final outcomes.

F Historical Metadata Experiments and Human Review

The earlier pipeline uses Amazon Reviews'23 product metadata, constructed queries, and a mixture of rule and model-generated relevance. Its early 48K report and later fixed100K acceptance study are distinct evaluations. The earlier report quotes BM25 NDCG 0.5515 and XGBoost 0.6681 on 9,600 queries, a 21.15% relative difference; the complete primary package is missing. A separate local development result is approximately 20.55%, which must not be substituted as the same run. The later auditable fixed100K study gives BM25 0.537934 and upgraded XGBoost 0.640150, or 19.00%.

The historical Week4 retriever develops beyond a simple SBERT/CLIP baseline: it includes BGE text representations and a cross-encoder-teacher projection/distillation stage. In the selected configuration, the parameter called alpha is zero in a target interpolation $(1-\alpha)$ teacher $+\,\alpha$ label; zero therefore means a pure teacher target, not no distillation. The selected multimodal Recall@50 is 0.612777 versus raw SBERT 0.494133, while strong text reaches 0.618449 and the matched image-off variant 0.630007. Index search P95 of 5.93 ms and loaded encoding-plus-search of 52.27 ms have different boundaries.

Historical Week5 uses a BGE cross-encoder prior with a 43-feature residual LambdaMART, reaching final NDCG 0.707681, 10.55% above the later Week2 baseline. The CE-only point estimate is 0.709045. Ordinary tree/listwise controls and deletions were also evaluated, but neither the combined relative gain nor a favorable model name proves a residual contribution. This architecture is not the final human-judged system.

The first historical Week6 adapted-vector/LLM combination performs poorly: Cold NDCG is approximately 0.320325 versus a 0.499983 Week2 baseline on reused development queries. Later compensation, evidence, and Qwen reranking variants lead to a separately fixed 200-query prospective confirmation with 33,296 pairs. Its 95 Cold-eligible queries give Cold NDCG 0.480413 to 0.547784, or 14.02%, with absolute paired interval $[0.029887, 0.106904]$. The development 15.17% figure is not the final confirmation. The Cold proxy is no pre-2022 reviews, not verified zero clicks. Its role cannot be identified with the current product-ID Cold split.

The historical CLIP LoRA pilot and later A100 study are also retained as research experiments. They improve some alignment/fixed-pool measurements, but the later full-catalog evaluation contains 8,183 unjudged Top50 pairs and no supported catalog NDCG. They do not establish an adopted final retrieval improvement. Likewise, the historical content-only insertion demo grows a subset index

from 26,419 to 26,420 products and succeeds in self-vector/title-query lookup; this is functional evidence, not a quality or latency evaluation of insertion into the complete final ESCI service.

The human-review sequence begins with title-only inspection, followed by review with the fuller content available to the model judge and clarified ambiguous entries. The final 99 assessable pairs provide agreement, not a new query-complete human relevance benchmark. The selected sample, annotation lineage, and absence of independent visual judgments limit interpretation. Raw historical prompt-ablation and manual F1 acceptance evidence is incomplete; implemented query processing is not promoted to a verified final LLM-rewrite contribution. The current ESCI pipeline disables rewrite. The separate same-architecture metadata reconstruction has code and local checks but no real complete encoding/training run, so it contributes no controlled outcome table.

G Engineering Protocol, Compatibility, and Deployment Scope

G.1 Timing Boundaries and Regression Gates

Boundary A starts with cached query vectors and measures exact retrieval, candidate mapping, 28-feature construction, scaling, ListNet, and sorting. Boundary B adds genuine query encoding. Boundary C additionally includes HTTP handling, IPC, and serialization. Cached replay HTTP is a separate functional/performance mode and is not relabeled as B or C for arbitrary queries. All timings retain their original hardware, thread, warm-up, and cache conditions. CPU/GPU synchronization and separate stage timers are required where GPU work is measured; the accepted optimized core is CPU work in a GPU-equipped runtime.

The A2 core uses one aligned product-key cache and the same FAISS index, with no reduction in Top100 or altered score formula. Twenty quick and 300 repeated paired checks compare FAISS rows/scores, product keys, raw/scaled 28-column features, ListNet scores, and complete order. Later same-CPU A0/A2/P3 checks preserve equality, while a different CPU can change last-bit floating-point values. The early cross-CPU failure has matching candidate identities/order but different cosine-derived features and scores, so it cannot be called historical bitwise replay.

The specific `document_length` diagnostic finds 20 differing values among 2,000 candidate positions at approximately 10^{-7} scale. The formula is $\log(1 + \ell_p)$ from frozen float32 retained-vocabulary lengths; text source and tokenization must be checked before attributing a difference to content. The exact instruction-level CPU/logarithm cause remains unresolved. No score rounding, forced unsupported SIMD, tolerance relaxation hiding a changed rank, or model retraining is used to claim equivalence.

G.2 Online Encoder Compatibility

The live worker uses the frozen SBERT source and a pinned CLIP text projection with the original input construction for each branch. It accepts arbitrary text, not query IDs. A separate CPU worker environment uses Torch 2.2.2 and NumPy 1.26.4, while the core environment retains NumPy 2.5.3 and its frozen numeric dependencies. Isolation avoids changing the accepted core runtime merely to load the text encoders. Tokenizers/checkpoints are loaded once. A batch interface exists for up to 16 strings, but the actual service uses single-query calls; no dynamic cross-request batching result is claimed.

Table 22: Limited historical human review of sampled silver-labeled pairs.

Evidence shown	Pairs	Assessable	Exact	Exact rate	Binary	Binary rate
Original title review	100	100	43	43.00%	63	63.00%
Frozen full-text review	100	99	50	50.51%	70	70.71%

Notes. The full-text review reuses the same original stratified sample and human; prior title scores were visible and prior assisted review history is documented. Counts are unweighted agreement among assessable pairs. They are not ranking metrics or an independent visual relevance study. No query has a fully human-judged candidate pool in this sample; human-gold ranking gains cannot be computed.

The accepted 100-query compatibility gate has SBERT maximum absolute error 1.657754×10^{-7} and minimum cosine 0.999999999999515; CLIP values are 3.278255×10^{-7} and 0.9999999999992293. Norms and fused inputs are checked. Top100 sets agree for all 100 queries, raw candidate orders for 95, final Top10 orders for all 100, Top50 sets for all 100, and complete ranks for 97. The earlier probe with five complete-rank changes is a separate record. Agreement on these inputs supports the approved demo, not a guarantee of identical rankings for every future query or proof of the original SQID revision.

G.3 Local Repeated Load Procedure

The frozen local dispatcher has one compute slot, four numerical threads, capacity and maximum waiters 64, a 9.5-second queue deadline, a 10-second response deadline, and an 8,192-product LRU. Its key is operation type, exact query text (or replay ID), and requested TopK. Broad1000 has 1,000 unique keys; Hot64 has 64. Broad keys remain leased after HTTP timeout until a completion trace and a later idle snapshot establish safe reuse. Missing completion information does not make an unfinished computation disappear.

1. Start a fresh service and encoder worker for each repeat. Record CPU, numerical threads, asset identities and startup checks. Startup's core/encoder regressions are excluded from timing samples.
2. Load and warm models, then clear the product LRU before an empty-cache comparison. The common A0/A2 set contains 200 fixed queries. Each implementation starts empty; AB/BA order is fixed before running.
3. For a workload-conditioned scene, clear LRU contents and hit/miss counters again. Execute its fixed query list twice, recording both pass times, query counts and cache transitions, then begin measurement. Hot/Broad states are independently prepared. B/C execute real encoders on every pass and measurement; persistent query-answer and query-vector caches are disabled.
4. For Burst64, prepare 64 distinct queries per wave, wait on one gate and release together. Record actual send spread, every response/error and backend drain. Wait two seconds only after draining, then start the next wave; run 20 waves without retries.
5. For Sustained64, run 64 closed-loop clients with one unfinished HTTP request per client and think time zero. Stop issuing after 600 seconds, then drain all started work. Sample every 30 seconds. If leased/quarantined keys exhaust supply, preserve the actual issuing window and remaining no-load interval as an incomplete planned load.
6. Run the fixed 8/16/32-client, 180-second closed-loop scenes; execute 96-request overload and ten serial recovery requests. Each scene retains its own cache preparation and failures. Stop and drain before restarting workers for the next repeat.
7. Compute per-repeat Mean/P50/P95/P99/Max and the under-200 ms fraction. Average repeat means with sample SD; recompute pooled percentiles from homogeneous raw samples. Repeated inputs and contiguous time blocks are not independent new queries. Report success, all-status completion, coalescing and supply separately. Do not add A/B/C values or infer network-only cost by subtraction.

The broader run has four serial repeats, 57 measured scene observations and 69,036 terminal HTTP records. Repeat 4 stops during the next scene's conditioning, not through a measured model mismatch; repeat 5 is absent. The interrupted plan is not silently redefined as a successful five-repeat protocol. Completed scene summaries use actual repeat counts. Both failed Broad sustained observations remain visible, while equal-load aggregates use the two completed 600-second observations. Descriptive repeat-level bootstrap intervals use 5,000 draws with seed 20261029 and actual available repeats; they are not the original five-repeat certification intervals. The main conclusions do not depend on request-level independence assumptions or on crossing hardware boundaries.

Recorded startup ranges from 53.55 to 135.26 seconds and includes checks, not only weight deserialization. Sampled API peak RSS ranges approximately 4.78–4.85 GiB, with encoder startup RSS approximately 0.61 GiB. RSS is not exclusive physical memory. Per-30-second CPU uses process user-plus-system deltas and one logical core as 100%; cache/queue samples, available RAM and swap are retained in the accompanying CSVs. Resource correlation alone does not identify the cause of slower repeats.

G.4 Earlier Finite-Workload Measurements

Table 23 and Figure 8 preserve an earlier local session with a small warmed working set. The 640/640 burst and 1,413/1,413 sustained successes are valid for those finite workloads; they do not replace the subsequent broader repeated evidence or establish universal 64-client reliability. The subsequent run has one coalesced successful Hot response, rather than the earlier distinct-query tests' zero coalescing.

G.5 Functional and Deployment Scope

The backend preserves the original feature/scaler/ranking contract. Existing automated tests cover input validation, TopK, readiness, authorization, bounded queues and recovery. The English interface reads real catalog fields through FastAPI; replay credentials remain server-side. No artificial prices, ratings or review counts are displayed. The approved live encoder criterion is compatibility of vectors and returned results, not recovered exact historical SQID identity.

Table 23: Earlier local HTTP tests with finite warmed working sets; not pooled with the repeated local experiment.

Online HTTP workload	Success / sent	Mean (ms)	P95 (ms)	Max (ms)	Boundary
New-query sequential	30/30	155.2	226.4	325.7	10 distinct, 3 repeats
Warmed reference	96/96	92.5	108.9	215.2	96 distinct sequential
Ten 64-request bursts	640/640	2955.8	5516.5	6216.6	64 distinct per burst
120 s, 64 clients	1413/1413	5561.9	6024.7	6108.2	64 distinct working set
96-request overload	68/96	3201.4	5877.4	6252.5	28 explicit HTTP 429
Recovery	3/3	100.4	107.0	107.3	After overload

Resource / rate	Recorded value
Startup	64.273 s
Peak API + encoder RSS	5.191 GiB
Sustained success rate, including drain	11.248 requests/s

Notes. Latency summaries use successful responses only (68 for overload). Loopback HTTP includes live dual text encoding, IPC, retrieval, features, ranking and serialization; product images, browser rendering and network deployment are excluded. New-query inputs are absent from the frozen cache. Later queue tests use one Mac CPU compute slot, a warmed product-token cache of 8,192 entries and no retries; all successful response signatures match. Queue capacity is 64 with 9.5 s queue and 10 s response limits. RSS sums can double-count shared pages. Earlier versions retain a 1/640 burst ReadError and 11/1,174 sustained timeouts. These are finite workloads, not a production SLA.

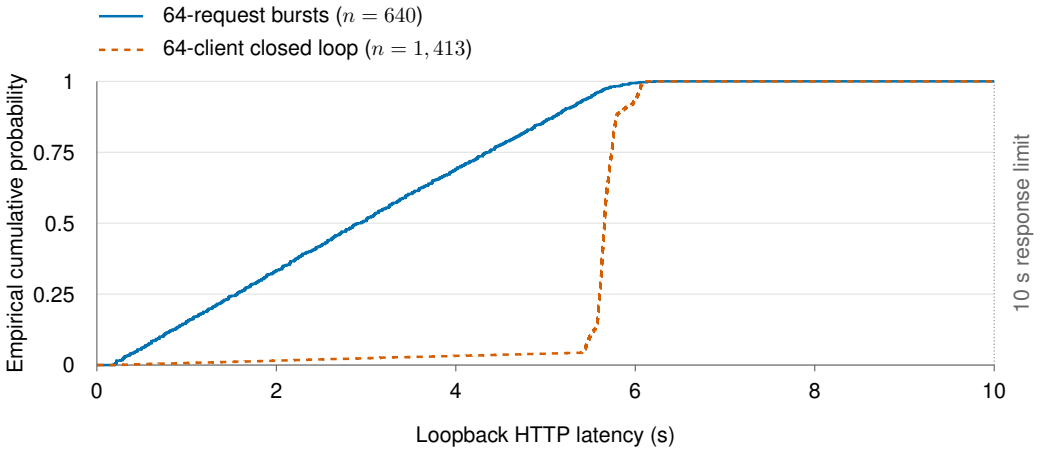


Figure 8: Earlier finite-workload HTTP distributions. This historical same-session figure is retained for context; the broader repeated tests in the main text include failures and incomplete issuing windows.

The Colab internal service worked, but external proxy browser acceptance was incomplete; the later local demo has browser evidence. No public API is deployed. An older Dockerfile lacks an accepted final build/launch record. The Week8 source package excludes weights, raw data, relevance labels and private runtime credentials. Source packaging and documentation do not constitute a new cross-machine runtime certification. Slides and demonstration video are deferred at the user's request.

H Archived error examples

Table 24 reproduces the existing selector's five largest improvements and five largest degradations for the final ListNet versus original Week2 comparison. Deltas are saved per-query Warm-target NDCG@10 differences (ListNet minus Week2), not percentage changes. Complete competitor positions are retained. These extreme cases are illustrative, not representative; they do not estimate failure frequencies or establish causes.

E/S/C/I denote the released Exact/Substitute/Complement/Irrelevant judgments; "?" denotes an unknown saved label. A displayed label does not identify whether that product is in the Warm metric target. An absent saved top-ten entry is written > 10; its exact lower rank is not exposed. Product titles below are excerpts. Labels are preserved even when they appear unexpected from the title, and no rejudging or metric recomputation was performed. The accompanying `evidence/error_case_notes.json` indexes all 60 archived cases and preserves complete saved titles.

Table 24: Archived extreme cases for ListNet versus original Week2 on the fixed judged pool (Warm target). Each product entry gives ID, released label and Week2 → ListNet position.

Query ID and text	Δ	Selected products: ID, label, old → new rank; title excerpt	Bounded observation
Five improvements			
23410 cartera de guchi	+0.8401	B0757ZLMLZ (I; $1 \rightarrow > 10$): Gucci Bloom for Women Eau de Parfum Spray… B077YC45WB (E; $> 10 \rightarrow 1$): Gucci GG Original Canvas With Grained Leather Trim Bifold Wallet…	Wallet/bag titles rise above perfume; no inferred spelling-correction mechanism.
73209 newborn baby boy bunjee	+0.7162	B000WJLKQM (E; $9 \rightarrow 1$): Graco Doorway Bumper Jumper, Little Jungle B07XFPD46L (I; $2 \rightarrow 2$): Cindear Infant Baby Boys Girls First Walker Shoes…	Jumper rises; a boot remains second. The intended meaning of "bunjee" stays uncertain.
100597 tangent lost the	+0.7016	B07X4TQSW5 (E; $> 10 \rightarrow 1$): Lost Then Found 1910089370 (I; $1 \rightarrow 3$): A Tragedy of Errors: The Story of Grace Murray…	Short, ambiguous query; the intended entity cannot be resolved from titles.
29049 cordless wall hanging fountains	+0.6782	B07K6JLP2C (E; $9 \rightarrow 1$): 5-Tier Relaxation Water Fountain with Lights… B071CTXG5Q (I; $1 \rightarrow 8$): 4-Pack Solar Powered Mason Jar Lights…	Fountain replaces decorative lights; cordless/wall-hanging attributes remain unverified.
26826 cleveland big and tall	+0.6382	B07TZLFF73 (E; $> 10 \rightarrow 1$): Vintage Guardian of Traffic Bridge Cleveland T-Shirt B087J9Z15T (I; $1 \rightarrow 9$): NFL Cleveland Browns Team Jersey Bodysuit…	T-shirt versus baby bodysuit; big-and-tall sizing is not visible.
Five degradations			
70106 modern ceiling fan without light	-0.5197	B09BF1JJ9K (I; $> 10 \rightarrow 1$): CHANFOK Ceiling fan with Light… B07YBRPL4G (E; $1 \rightarrow 4$): Westinghouse Lighting 7224200 Comet Indoor Ceiling Fan with Light…	Negated attribute. Both shown titles say with light, but released labels differ.

Continued on next page

Table 24 (continued)

Query ID and text	Δ	Selected products: ID, label, old → new rank; title excerpt	Bounded observation
103771 toddler sports bedding	-0.4262	B000R422LA (I; $> 10 \rightarrow 1$): Mk Collection Bedspread Boys Sport Football Basketball Baseball New (Twin) B06XJJBYXV (E; $1 \rightarrow 9$): Carter's 4-Piece Toddler Set…	Twin bedspread displaces toddler set; visible age/bed-size distinction.
59544 ladder self closing for pool	-0.4109	B002ZJ8H1I (I; $6 \rightarrow 1$): In-Ground Pool Safety Fence Drill Guide Deposit B00ITPPPPI (E; $1 \rightarrow 2$): Blue Wave NE115T Neptune A-Frame Entry System for Above Ground Pool…	Related pool safety equipment displaces entry system; self-closing not shown.
41911 food freeze dryer machine not dehydrator	-0.3969	B08JZ45W7W (I; $8 \rightarrow 1$): Dehyber Food dehydrator… B07SQMVLFH (E; $1 \rightarrow> 10$): Brod & Taylor SAHARA Folding Food Dehydrator…	Both shown titles say dehydrator despite query exclusion; labels differ.
65676 mar'villa	-0.3936	B086N39CX5 (?; $2 \rightarrow 1$): Mar'villa UV-C Phone Sanitizer & Wireless Charging Stand… B087JM7KB1 (E; $1 \rightarrow 3$): Warrior Wand The Best Portable Medical Grade UV-C Sanitizer Wand…	Brand-matching title rises but is unknown. Metric decline need not mean worse intent match.

Catalog inspection boundary. The saved auxiliary contrast is RRF Top100 followed by ListNet (B) versus calibrated Fusion Top100 followed by the same ListNet (A). Its Cold known-positive Recall@50 deltas cannot be fully explained by the saved top ten. For query 112403, "woman ski pants", the saved A and B top ten are identical even though the recorded Cold recall delta is -1.0. The displayed E labels do not identify Cold membership, and unknown results must remain unknown.

I Use of AI Assistance

AI tools were used during the project to support code scaffolding, error diagnosis and debugging, code review, and the organization of research ideas and experimental workflows. They also assisted with documentation and language revision. The author remains responsible for the final methodological decisions, interpretation of results, and accuracy of the report.